
Contents

CV of Rohit Chaturvedi (Individual Consultant) 3

A. DIAGNOSTIC & POLICY RELATED ASSIGNMENTS 4

A. OTHER ASSIGNMENTS.....10

B. PORTS & SHIPPING (PPP & TRANSACTION ADVISORY).....29

C. PORTS & SHIPPING (TRAFFIC & OTHERS)32

D. SHIPYARD44

E. ROADS.....47

F. AVIATION:.....54

G. LOGISTICS:56

H. CONTAINERS66

I. RAILWAYS:68

J. FINANCIAL SERVICES:72

K. PPP & OTHERS85

CV of Rohit Chaturvedi (Individual Consultant)

1.	Proposed Position	Individual Consultant			
2.	Name of Firm	RC & Co.			
3.	Name of Staff	Rohit Chaturvedi			
4.	Date of Birth	8 th April 1975	Nationality		Indian
5.	Education Qualification				
	Degree Obtained	Institution/University		Year	
	PGDM(MBA)	Indian Institute of Management Bangalore (IIMB), Bengaluru, India		2001-2003	
	Bachelor of Law (LLB)	University of Rajasthan, Jaipur, India		2017-2020	
	Bachelor of Engineering, Kumaun Engineering College	Nainital University, Nainital, Uttarakhand, India		1993-97	
6.	Membership of Professional Associations	Member, National Logistics Committee, Federation of Indian Chambers of Commerce & Industry			
7.	Other Training	Machine Learning by Coursera Course Certificates			
8.	Countries of Work Experience	India, Namibia, Tanzania, Sierra Leone, Indonesia, Bangladesh, Singapore, Tunisia , Egypt			
9.	Languages	Language	Speaking	Reading	Writing
		English	Excellent	Excellent	Excellent
		Hindi	Excellent	Excellent	Excellent
10.	Employment Record				
	From	To	Employer		Positions Held
	July 2016	Current	RC & Co. Kitzo Management Advisory LLP		Founder
	April 2012	May 2016	CRISIL Infrastructure Advisory, a division of CRISIL (A Standard & Poor Company)		Director
	November 2010	March 2012			Head – Transport & Logistics
	June 2010	October 2010			Director - Projects
	April 2009	May 2010	i-maritime Consultancy Private Limited		Head (Project Delivery)
	June 2008	March 2009			Principal Consultant
	March 2008	June 2008	Jasper Infotech Pvt. Ltd.		Member of Start-up Team
	March 2007	March 2008	KPMG Advisory Services		Senior Consultant
	November 2005	March 2007	AF Ferguson & Company (Erstwhile a member firm and subsequently merged with Deloitte)		Senior Consultant
	July 2003	Nov 2005	Indian Oil Corporation Ltd		State Head, Fleet Marketing Officer (Delhi & Haryana)

	October, 1998	June 2001	Videsh Sanchar Nigam Limited	Senior Engineer, Internet R&D
11.	Work Undertaken that Best Illustrates Capability to Handle the Tasks Assigned			
	A. DIAGNOSTIC & POLICY RELATED ASSIGNMENTS 1. Name of the Assignment: Implementation Review of Short-Term Action Plan of NEPAD Year: September 2009 – December 2011 Location: Africa Client: African Development Bank Main project features: The study involved reviewing the progress and implementation issues in the New Partnership for Africa's Development (NEPAD) short-term action plan (STAP) involving 120 key infrastructure projects spread across Africa. The major infrastructure sectors adopted by the NEPAD infrastructure programme are transport, energy, information and communications technology (ICT) and trans-boundary water resource management. The 2009 review of the NEPAD STAP projects includes 103 projects spread across seven regional economic communities (REC). The STAP project portfolio consisted of four types of projects: investment projects (39%), capacity-building projects (14%), studies (14%), and facilitation projects (17%). The STAP project portfolio consisted of projects from four sectors: transport (46%), ICT (17%), energy (26%), and water (11%). The projects have also been classified across the RECs. The Arab Maghreb Union (AMU) and the Common Market for East and Southern Africa (COMESA) supervised 14 and 11 projects, respectively. The East African Community (EAC) and the Economic Community of Central African States (ECCAS) were coordinating 13 and 31 STAP projects, respectively. The Economic Community of West African States (ECOWAS) and the Inter-governmental Authority on Development (IGAD) monitored 42 and 13 projects, respectively, while the Southern African Development Community (SADC) was in charge of 30 projects. The review exercise comprised the following tasks: • Overviewing major issues arising out of the last STAP implementation review • Review and analysis of project implementation status based on type, sector, and REC • Assessment or recommendation of the option of prioritizing the projects/programmes • Highlighting the areas of STAP, where the program for Infrastructure Development in Africa (PIDA) can intervene and take the projects forward • Identification of major challenges • Recommendations for going forward Position held: Project Leader Activities performed: Prepared the framework of the study to be carried out by the team, was involved in all the client interactions presentations, supervision, reviewed the output quality and acted as the key interface for the client 2. Name of the assignment: Consultant for preparation of comprehensive MARTIME policy incorporating all related and ancillary activities and providing transaction advisory services to MMB Year: March 2015 – May 2016			

Location: India

Client: Maharashtra Maritime Board

Main Project Features:

The government of Maharashtra published its port policy in 1996 and thereafter improved upon it in the year 2000, 2002 and 2010 in order to make it development-friendly. It also published its Captive Jetty Policy in 1995 and improved upon in the year 2005 and notified Multipurpose Jetty Policy in 2005. Various projects have been approved under these policies by the Board, some of which have been commissioned while some are undergoing development. A number of ancillary activities like marinas, water sports, shipyards and ship breaking etc. are currently being handled on a case-to-case basis. Therefore, the Maharashtra Maritime Board has decided to prepare a Comprehensive Port Policy which includes various ancillary-related activities.

Position Held: Project Leader

Activities Performed: Supervision, review of output quality and creating key interface for the client.

3. Name of the assignment: PPP institutional reform capacity building and pipeline development support

Year: November 2013 to September 2014

Location: Sierra Leone

Client: The World Bank

Main Project Features: The Government of Sierra Leone (GoSL) planned the use of public-private partnerships (PPPs) as a mechanism for procuring and financing infrastructure projects and services in the public sector. To achieve this goal, the World Bank Group (WBG) is supporting the recently established PPP Unit under the Office of the President, GoSL.

The WBG engaged the services of CRIS as consultant to help in the initial stages of PPP pipeline development

Position held: Project Leader

Activities performed:

- PPP Project development advisory support focusing on assessing and developing the GOSLs current pipeline of PPP projects:
- Reviewed and discussed the current pipeline of projects in the energy and other infrastructure sectors with the GOSL, to assess the feasibility of private sector participation as well as provide an assessment of the technical, financial, operational and management situation (including pros and cons) of potential transactions
- Identification of potential greenfield projects:
- The report also included recommendations on feasible greenfield projects for the short and medium-term
- A short options paper on possible GoSL's models of engaging concessionaires on energy
- Drafting of a short options paper which will outline the possible GoSL models of engaging concessionaires on government acting as an off taker from concessionaires that build power plants, government having equity participation in SPV vehicles for generation, government as an owner and selling to companies

4. Name of the assignment: Assistance in operationalizing Namibia's PPP Policy

Year: June 2013 – July 2014

Location: Namibia

Client: Ministry of Finance Namibia

Main Project Features: The Government of Namibia has taken the first steps towards attracting private investments in the areas of economic, social, municipal, or social assets by way of developing a national PPP policy, as per whose provisions, the Ministry of Finance (MoF) is expected to take a leading role in enablement and approval of PPP projects in the country. With this backdrop, MoF appointed CRIS to assist them on various activities related to operationalizing the PPP policy.

Position Held: Project Leader

Activities Performed: The key activities envisaged under this engagement are:

- Legislative review and law creation to enable PPPs
- Development of a 'review and approval process' for annuity PPP projects to ensure fiscal soundness
- Preparation of guidance material for all stages of the PPP process
- Assistance in organization of the PPP unit under the MoF
- Capacity building: design and delivery of agreed PPPs related training programs
- Identification of two pilot PPP projects, which can be further taken up for development and PPP procurement
- Stakeholder consultations
- Preparatory works for approval of the PPP Bill
- Taking the Bill through various stages of approval process

5. Name of the Assignment: PPP Pipeline and Viability Gap Mechanism Assessment

Year: August 2012 - Dec 2012

Location: Ghana

Client: World Bank

Main Project Features: The Ghanaian and Kenyan governments have proposed 11-12 projects that constitute the 'first mover projects' (FMP) under their respective PPP policies. The initial assessment of the suitability of these projects for PPP financing is being carried out. On the basis of the financing requirement assessments (equity, debt, guarantees, etc., and other government support), the bank will be: (i) providing guidance to the governments on how best to structure financing instruments, including VGM options, and (ii) providing funding support to some of these options under Phase II of the programme.

Position Held: Project Leader

Activities Performed: The scope of services provided by CRIS included:

- Preparing the preliminary VGM profile for each of the FMPs in Ghana and Kenya;

- Compilation of international approaches\methodologies to VGM, pros and cons, and performance to date (e.g., IUK, South Africa, India, Chile, Brazil; Malaysia, South Korea, New Zealand, Canada) building also on work done to date by World Bank;
- Recommendations on design options for as to the most efficient approaches for a VGM with specific attention to ongoing integrated financial management information systems (IFMIS) underway in the respective countries to insure effective VGM integration into overall budget systems to allow for effective\efficient recording of VGM allocations in main estimates and regular reporting;
- Participation in the September 10-21 mission to Ghana and Kenya;
- Assistance on organisation of South-South Practitioner Workshop to be hosted in Nairobi, including background presentations

6. Name of the Assignment: Assessment of coastal cargo in Gujarat

Year: November 2014 – July 2015

Location: Gujarat, India

Client: Gujarat Infrastructure Development Board

Main Project Features:

Gujarat Maritime Board (GMB) intends to assess the potential of coastal cargo movement in the state of Gujarat. The Authority, through a tendering process taken up by Gujarat Infrastructure Development Corporation (GIDB), has selected CRISIL Risk & Infrastructure Solutions (CRIS) to prepare a report for assessment of coastal cargo potential in Gujarat. The scope of work entails assessment of interstate movement of commodities; understanding the coastal cargo movement in India and Gujarat; assessing possible modal shifts and therefore potential of future coastal movement of cargo in Gujarat, apart from evaluating other factors such as regulations impacting coastal cargo and providing a list of viable routes/commodities. The study also provided policy inputs to the project authorities .

Position Held: Port sector specialist

Activities Performed: Supervision, presentations to client

7. Name of the assignment: Advisory services for product library development and resultant policies, processes and manuals

Year: March 2013 to July 2013

Location: Indonesia

Client: PT Sarana Multi Infrastruktur (Persero) Tbk

Main Project Features: PT Sarana Multi Infrastruktur (Persero) was established in 2009 to accelerate infrastructure development in Indonesia. PT SMI focused on funding the commercially-feasible infrastructure projects through debt, equity, and mezzanine instruments to address the constraints on the flow of investment in infrastructure. In order to provide financing to infrastructure projects in Indonesia on reasonable terms, the PT SMI management was looking to develop an effective product library and having a suitable enabling environment in terms of policies, processes, and manuals to support the same.

PT SMI thus appointed CRIS, India, to provide advisory services in this regard.

Position held: Project Leader

Activities performed:

- Review and improvement of product library
- Development of financing and investment policies
- Review and improvement of business processes
- Development of credit and investment process manuals
- Conducting a workshop on effective implementation

8. Name of the Assignment: Evaluation of Business Plan for NEIA Trust

Year: December 2012 to June 2013

Location: India

Client: Export Credit Guarantee Corporation (ECGC) of India

Main Project Features: The National Export Insurance Account (NEIA) Trust is a public trust set up in 2006 by the Government of India and administered by the ECGC. The trust provides export credit insurance support to facilitate medium- and long-term exports to difficult markets. The trust also provides covers to banks for buyer's credit transactions, which facilitates foreign buyers to pay for project exports from India. ECGC had appointed CRIS to evaluate its business plan.

Position held: Project Leader

Activities Performed: The scope of work for the current study included:

- Performance evaluation of the NEIA Trust, since its inception in 2006
- Scope of expanding the NEIA Trust's coverage beyond medium- and long-term business
- Specific measures to boost the exports to specified countries to correct India's trade imbalance, through the trust.

9. Name of the assignment: Developing Tool Kit(s) for Improving PPP Decision-Making Processes

Year: August 2009 to July 2011

Location: New Delhi, India

Client: The World Bank

Main Project Features: The objective of the proposed assignment was to develop a web-based online 'PPP Toolkit' to facilitate identification, assessment, development, procurement, and monitoring of public private partnership (PPP) projects, with a focus on 4-5 selected infrastructure sectors— roads, ports, urban transport like bus rapid transit system (BRTS), water supply and sanitation, and solid waste management.

These toolkits—both generic as well as sector-specific variants—provided relevant, practical, and best practice guidance to officials at the national, state, and city/local governments, state-owned enterprises, and other agencies responsible for developing and monitoring PPPs in the selected sectors. Accordingly, the toolkits had been developed based on an assessment of performance of a representative set of PPPs that were already awarded and were in operation, so as to build upon the lessons from the past experience. Additionally, they incorporated international best practices by drawing upon toolkits that were already developed by the World Bank/Public-Private Infrastructure Advisory Facility (PPIAF)/other agencies. These toolkits were expected to be particularly useful for lagging states and regions with huge infrastructure deficit and low capacity to conceptualize and develop PPP projects. The toolkit also contained case studies of 15 landmark PPP projects \.

Position held: Project Leader
Activities performed:

- Reviewed selected PPP projects across identified sectors as case studies to identify learnings that could be used for development of toolkits
- Provided necessary inputs for the development of web-based online toolkits
- Prepared MS Excel models
- Prepared user manuals, and organize and conduct stakeholders’ meetings

10. Name of the assignment: Study on development of ship repair and ship building strategy for EGYPT and development of policy

Year: 2009 to 2010

Location: Egypt

Client: Government of Egypt (GOE)

Main project features: Managed the study for the client with the overall objective of suggesting a policy framework to facilitate development of the ship repair and shipbuilding sector in Egypt and suggesting measures to increase international competitiveness. The specific objectives included the following:

- Global review of the sector
- Assessed efficiency of shipyards in Egypt and performed a competitive analysis
- Identifying needed improvements
- Formulating a development strategy for the industry
- Preparing an action plan for the ship repair and ship building development strategy

Position held: Project Director & Team Leader

Activities Performed: Supervision, review of quality of output, key interface for the client

A. OTHER ASSIGNMENTS

11. Procurement of Consultancy Services for provision of PPP Transaction Advisory Services for “Detailed Business Case Model for Terminal 2 & Terminal 3 of bay. Terminal Project under CPA”

Year: December 2021 – June. 2022

Location: Bangladesh

Client: Ernst & Young LLP (EY)

Main features: The study included following key streams of work:

- 1) PPP Transaction Structuring
 - a) Assessing the Existing Tariff Structure of CPA
 - b) Applicable Fiscal or Special Incentives for PPP
 - c) Identification of PPP guidelines for incentives
 - d) Listing out the Base Assumptions of Feasibility Study
 - e) Financial Model (Considering DBFOT)
 - f) Key Components of Financial Model
 - g) Scenario Analysis
- 2) Project Risk assessment:
 - h) Risk definition & identification
 - i) Risk Assessment
 - j) Risk Allocation
 - k) Risk Mitigation
- 3) Heads of Terms of Concession Agreement
 - a) Parties, Roles & Responsibility of each party, Concession period, Tariff, Key Clauses,
 - b) Roles and Responsibility of G2G Partner
 - c) (The roles and responsibility shall be based on the identified Project Structure and the G2G agreement)
 - d) Roles and Responsibility of CPA & Other Governing Bodies
 - e) (The roles and responsibility shall be based on the identified Project Structure and the G2G agreement)
 - f) Setting of Key Commercial Terms and conditions of the Concession Agreement

Position held: Team Leader

Activities performed: Overseeing entire study with.

- a) Review of reports and other deliverables
- b) Guiding the team comprising various experts
- c) Interfacing with the client and making presentations

12. Due diligence of 1 million TEU capacity container terminal on the west coast of India

Year: December 2021 – June. 2022

Location: Maharashtra, India

Client: Bothra Shipping Limited

Main project features: The study included detailed analysis of container market in India and competitive scenario for Container Terminal:

- Macro analysis of Indian Economy & establishing statistical relationship with container volumes in India
- Estimate volumes for the said container terminal and projections for 5 years
- Assessment of capacity and bottlenecks based on marine, cargo handling, civil & connectivity infrastructure
- Quick financial statement with Free Cash Flow projections
- Calculation of NPV and IRR for valuation purposes
- Interaction with the target on regular basis
- Sensitivity analysis based on variation of volumes and capital expenditures
- Advising client on all incidental matters including legal, regulatory and commercial matter

Position held: Project Director

Activities performed: Worked on a module and directed the overall study with the client's team. Reviewed the output and undertook all the customer interactions.

13. Due diligence exercise for Container Freight Stations (CFS) in 4 largest container port locations of India

Year: October 2021 to November 2021

Location: Mumbai, Maharashtra, India

Client: Alvarez & Marsal, Unit 703 / 704, 7th Floor, Peninsula Tower A, Peninsula Corporate Park, Ganpatrao Kadam Marg,, Lower Parel, Ganpatrao Kadam Marg, Lower Parel West, Lower Parel, Mumbai, Maharashtra 400013

Main project features: The study included detailed analysis of container market in India and competitive scenario for Container Freight Stations (CFS)

- Macro analysis of Indian Economy & establishing statistical relationship with container volumes in India
- Micro analysis
 - Region wise origin/destination from/to port to different region
 - Share of each region to each port
 - Port's regional share of container volumes
 - Growth rate of each region based on macro indicators of each region
 - Analysis of modal changes in transportation and its impact
 - Estimate port volumes based on the above analysis
- Competitive scenario of each Container Freight Stations (CFS) in each of the port locations under study
- Analysis of each CFS in terms of past trends of volume and their locations viz-a-viz respective ports
- Review of DPD (Direct Port Delivery) policy for import volumes of container and its impact on CFS volumes
- Other development such a shift of LCL(Less than container load) , bonded warehouse etc,

- affecting competitive positioning of each CFS
- Policy review on development of CFS policies

Position held: Project Director

Activities performed: Worked on a module and directed the overall study. Reviewed the output and undertook all the customer interactions.

14. Valuation exercise for coal and fertilizer terminals at Kakinada Port on East Coast of India

Year: April 2021 to August 2021

Location: Kakinada, Andhra Pradesh, India

Client: Bothra Shipping Limited

Main project features: The study focused on valuing the terminals for coal and fertilizer at Kakinada, India. The study encompassed:

- Understanding the financial statement of the both the terminals
- Review tariffs at the terminals and assess the trend going forward
- Estimated the asset live of key assets from the perspective of useful life replacement requirement
- Cost estimates of assets both depreciated assets and new assets required in future
- Detailed operational cost estimates
- Cash flow projections and valuation based on Discounted Cash Flow (DCF) methodology
- Carried out valuation based on book value of assets and industry benchmark
- Review of recent transactions in the industry and carried out valuation from EBITDA (Earning Before Interest Tax Depreciation & Amortization) multiple methodology

Position held: Project Director

Activities performed: review of all the deliverables and provide continuous inputs and led client interactions

15. Market analysis, strategy and business plan development, and capacity building to promote low carbon development and modal shift to Inland Water Transport (IWT) in Bangladesh

Year: Feb 2018 to Aug 2018

Location: Dhaka, Bangladesh

Client: CPCS, 72 Chamberlain Avenue, Ottawa, Canada , K1S 1V9 , ottawa@cpcs.ca, www.cpcs.ca

Main project features: During the study, the team studied the logistics chain of industries especially garment industry in Bangladesh between Dhaka -Chittagong Corridor. The work involved :

- Logistics Supply Chain Mapping & Modal Shift Assessment
- Low Carbon Waterway Transport Strategy & Intervention Plan
- Pilot Program Business Plan for IWT Sector
- Implementation Support for Pilot Program

Position held: Inland Waterways Expert

Activities performed: As is Analysis of Logistics Chain, Current Barriers Recommendation on Optimal logistics chain

16. Name of the assignment: Devising a plan for decongesting Mumbai through alternative transport modes

Year: 2008 to 2010

Location: Mumbai, India

Client: Inland Waterways Authority of India (IWAI)

Main project features: During the study, the team:

- Assessed cargo and passenger traffic that could be attracted by inland waterways and coastal routes
- Assessed possibility of shifting traffic to alternative modes of transport under time and cost constraints
- Assessed viability of alternative transportation mechanism under the above backdrop
- Suggested implementing strategies including the public-private partnership (PPP) model to attract private investors
- Developed an organisation structure to support the initiative involving various stakeholders in the central and state governments

Position held: Project Director & Team Leader

Activities performed: Supervised preparation of plan for decongesting Mumbai through alternative modes of logistics plan

17. Name of the assignment: Feasibility study for establishing commercial viability of movement of containers on riverine routes of national waterways

Year: August 2012 – March 2013MADC

Location: India

Client: Container Corporation of India Ltd. (CONCOR)

Main Project Features: CONCOR, one of the largest logistics players in India providing logistic solutions to India's domestic and international trade, wished to undertake a feasibility study for establishing commercial viability of movement of containers on riverine routes of national waterways. These riverine routes are the six national waterways in India. CONCOR appointed CRIS to conduct a detailed feasibility study and recommend a business plan for container movement on national waterways.

Position Held: Project Leader

Activities Performed: The scope of services provided by CRIS included:

- Project overview
 - Inland waterways in India
 - Containerisation in India
 - Environmental analysis

- Value chain analysis
- Demand potential analysis
 - Identification of hinterland along all the NWs
 - Identification of major industries and clusters and consumption centres along the NWs
 - Traffic profile of major industries commodities and clusters
 - Estimation of the overall potential traffic movement from major modes along the NWs
- Supply-side analysis
 - Total logistics cost analysis
 - Comparative total logistics time
 - Qualitative analysis across competing nodes
 - Potential share across competing modes
- Capital and operational expenditure estimation
- Financial feasibility
 - Financial statements – Balance sheet, P&L, and cash flow statements
 - Key indicators (NPV, IRR, DSCR)
 - Sensitivity analysis
 - Preparation of a bankable business plan to enable strategy formulation and investment decision by CONCOR

18. Name of the assignment: Consultancy services for the preparation of concession agreement for Kasanga port in Rukwa region

Year: June 2011- July 2012

Location: Kasanga

Client: Tanzania Ports Authority

Main project features: During the study, the team:

- Prepared documents for obtaining approval for upfront tariff
- Evaluated applications received for qualification and provided assistance in shortlisting parties
- Prepared and finalised tender documents, namely, Request for Proposal (RFP) and Draft Concession Agreement (DCA)
- Evaluated the financial proposals received and assisting in finalising the successful bidder
- Assisted in signing the concession agreement
- Assisted in the selection and appointment of an independent engineer for the project

Position held: Project Leader

Activities Performed: Supervision, review of quality of output, key interface for the client

19. Name of the assignment: Development of Ro-Ro services in Gulf of Kutch, Gujarat, India

Year: June 2016 – Feb 2017

Location: India

Client: Gujarat Maritime Board

Main Project Features:

Gujarat Maritime Board is carrying out a feasibility study to develop Ro-Ro jetties in the Gulf of Kutch in Gujarat, India. As a traffic expert, the role was to undertake the traffic study and estimate the number of passengers and vehicles that can be attracted to use the proposed infrastructure. Also based on the inputs received from the technical team various options and their financial feasibility was

analyzed. In the process, a detailed financial model was developed including tariff estimation, opex and capex , cost of capital etc. Most optimal option was suggested to the client.

Position Held: Traffic expert and financial modeler

Activities Performed: Approach and methodology, Supervision, review of output quality and key interface for the client.

20. **Name of the assignment:** Review consultant for Traffic study for 8 Ro-Ro terminals in Mumbai region

Year: June 2016 – March 2017

Location: India

Client: i-maritime Consultancy and Maharashtra Maritime Board(MMB)

Main Project Features:

The role was to suggest the methodology for traffic potential and review the outcome . As a part of study, competing modes of transport and likely shift to proposed Ro-Ro terminals was analyzed based on cost and time parameters. Study also included primary interactions with the potential users such passenger and commercial vehicles, user using public transport. The study was done for entire Greater Mumbai Region .

Position Held: Review Consultant

Activities Performed: Approach and methodology, Supervision, review of output quality and key interface for the client.

21. **Name of the assignment:** Consultant for preparation of comprehensive port policy incorporating all related and ancillary activities and providing transaction advisory services to MMB

Year: March 2015 – May 2016

Location: India

Client: Maharashtra Maritime Board

Main Project Features:

Government of Maharashtra published its port policy in 1996 and thereafter improved upon it in the year 2000, 2002 and 2010 in order to make it development-friendly. It also published its Captive Jetty Policy in 1995 and improved upon in the year 2005 and notified Multipurpose Jetty Policy in 2005. Various projects have been approved under these policies by the Board, some of which, have been commissioned while some are undergoing development. A number of ancillary activities like marinas, water sports, shipyards and ship breaking etc. are currently being handled on case-to-case basis. Therefore, the Maharashtra Maritime Board has decided to prepare a Comprehensive Port Policy which includes various ancillary-related activities.

Position Held: Project Leader

Activities Performed: Supervision, review of output quality and creating key interface for the client.

22. **Name of the assignment:** Traffic and feasibility assessment for Revdanda Port

Year: July 2015-November 2015

Location: Maharashtra, India

Client: Indo Energy International Limited

Main Project Features: Indo Energy International Limited (Client) is a private sector port infrastructure developer and provider of shipping services in India. It handles import and export of bulk cargoes like steam coal, iron ore, bauxite, limestone, sulphuretc and has an existing aggregate capacity of 5 MTPA at Sanegaon where it owns a 200m multipurpose jetty.

The client is developing a deep water port at Revdanda, about 40 nautical miles south of Mumbai harbour. The current plan is to construct two berths to bring capesize vessels for dry bulk and fifth generation container vessels alongside. Initial technical viability study and surveys have already been done while the Detail Project Report (DPR) is currently being prepared.

The primary objective of this study is to undertake a traffic assessment study, conduct financial analysis and determine phase-wise port development plan for Revdanda port. It will also include assessment of risk from a regulatory and environmental standpoint.

In this context, the client had requested CRISIL Risk and Infrastructure Solutions Limited (CRIS) to submit its proposal for undertaking the study.

Position Held: Project Leader

Activities Performed: The scope of work included:

- Undertaking Traffic Potential Study
 - Current status of the port
 - Traffic potential analysis
 - Revdanda port analysis
 - Vessel analysis and impact of dredging
 - Potential for additional developments in the port
- Financial analysis and phase-wise port development plan
- Regulatory and environment study

23. Name of the assignment: Coal study for Kakinada deep water port

Year: June 2015- Oct 2015

Location: Andhra Pradesh

Client: Bothra Shipping Services Private Limited

Main Project Features: The key objectives of the study are:

- To understand the impact of external environment for coal traffic at Kakinada Port in the wake of current and increased future demand of thermal coal in India
- The future potential will determine the business case for the choice of project viz.
 - Augmenting existing capacity by incremental changes OR
 - Develop full-fledged coal terminal with the capacity of 10-12 million tonnes, in addition to the existing Berth 5 with 8 MMTPA capacity

Position Held: Project Leader

Activities Performed: The scope of work included:

Module 1: Market potential for coal at Kakinada deep water port

- As-is analysis of logistics chain for existing customers
- Competitive analysis
- Mapping of existing customers versus potential (new) customers

- Key drivers to attract potential customers to Kakinada port
- Forecast of coal (both thermal and coking) potential for Kakinada port
- Key recommendations to realise the potential

Module 2: Financial feasibility assessment for coal terminal at Kakinada deep water port

24. Name of the assignment: Revalidation of Kattupalli Port Business Plan

Year: February 2015-April 2015

Location: Tamil Nadu

Client: Larsen and Toubro Limited

Main Project Features: Katupalli Port is **developed by one** of the most reputed Indian company L&T. The port has been recently acquired by Adani Group, the largest company in India in port's sector. The primary objective of this exercise was to undertake a revalidation of traffic assessment and projections made by CRISIL for Kattupalli Port in its previous report, in light of the developments in hinterland during the last 2 years.

Based on the initial discussions with the client, the scope of work shall be limited to:

- Revalidation of traffic estimates for containers, Ro-Ro, POL products, edible oil, barytes, and granite
- Fresh traffic study for thermal coal, coking coal and finished fertilisers
- Tariff benchmarking

Position Held: Project Leader

Activities Performed: The scope of work included:

- Revalidation of traffic estimates
- Traffic study for coal and fertilizers
- Tariff benchmarking

25. Name of the assignment:Traffic study and business plan for Kattupalli port

Year: May 2013 – June 2014

Location: India

Client: L&T Shipbuilding Limited

Main Project Features: Kattupalli International Container Terminal (KICT) is located near Chennai, adjacent to the Ennore port in Thiruvallur district of Tamil Nadu. It is being developed by L&T Shipbuilding Limited (LTSB), a joint venture between Larsen & Toubro (L&T) and Tamil Nadu Industrial Development Corporation (TIDCO). L&T Shipbuilding Limited (LTSB) has mandated CRIS to prepare a strategic business plan and conduct a traffic study for containers and other major cargo categories.

Position Held: Project Leader

Activities Performed:

The Scope of Work included the following modules:

- Assessment of traffic for containers and other major cargo categories:

This exercise included the following activities:

- Assessing the container traffic potential for the port of Kattupalli
- Identification and assessment of cargo traffic for the major non-containerised commodities that the port can attract
- Competition benchmarking for the port
- Tariff/pricing structure
- Preparation of a Strategic Business Plan:

This exercise included the following activities:

- Diagnostic evaluation of environment
- Strategic and action planning

26. Name of the assignment: Master plan for Gujarat ports

Year: February 2013 – October 2013

Location: India

Client: Haskoning India Pvt. Ltd.

Main Project Features: Haskoning India Private Limited (RHI) is developing master plan for three existing ports in Gujarat, owned by a major private port player. These ports are currently handling captive cargo and the port player is interested in developing facilities for handling third party cargo. RHI has invited CRISIL Infrastructure Advisory (CRISIL) to provide its services for undertaking traffic assessment and forecasting for these ports along with financial feasibility exercise. In addition, the scope would also include preparation of visioning exercise, including preparation of strategic plan and goals and identification of strategic levers to develop and compete effectively.

Position Held: Project Leader

Activities Performed: The Scope of work includes the following tasks:

- Traffic study covering captive potential, non-captive cargo, transshipment traffic
- Visioning and strategic advice
- Financial analysis
- Regulatory advice

As part of the traffic study, CRIS has assessed and projected traffic for six different types of commodities –coal, container, fertiliser, LNG, LPG and liquid traffic- along with the potential for a Ro-Ro terminal, separately for each of the three ports. To project the traffic for each of these commodities, traffic forecasting models were prepared taking into account the total logistic cost incurred to transport these commodities via various competing ports.

The next phase of the assignment would involve preparation of strategy for the short, medium and long-term development of ports followed by a financial viability assessment.

27. Name of the assignment: Feasibility analysis of transportation by sea route

Year: August 2012 - September 2012

Location: India

Client:CRH, one of the largest Cement players globally

Main Project Features: One of the largest cement players in Europe wishes to expand its presence in India inorganically. As a part of its overall India strategy, the company is now evaluating the acquisition of multiple assets in India, including two cement plants and a captive jetty. CRISIL Infrastructure Advisory was appointed as a consultant to assess the technical and commercial feasibility of logistics of cement and clinker by coastal shipping.

Position Held: Project Leader

Activities Performed: The scope of services provided by CRIS included:

- Assessment of scale and existing logistics operations
- Review of coastal shipping mode
 - Route assessment and feasibility
 - Voyage analysis
- Viability assessment and risk identification
 - Logistics cost analysis
 - Suggested options for the client

28. Name of the assignment: Business case for development of container berth

Year: May 2012 - December 2012

Location: Andhra Pradesh, India

Client: Kakinada Deep Water Ports

Main Project Features: A non-major port in India is currently expanding with the construction of three additional berths, one each with fully mechanised coal handling and fully mechanized fertiliser handling berths and a third being envisaged to be developed into a dedicated container berth. The client wishes to undertake the mechanisation and operation of the additional three berths at the port. CRIS is assisting the client by developing a business case for it and support in finding various options and alternatives for developing container terminals at the port.

Position Held: Project Leader

Activities Performed: The scope of work for this assignment includes preparation of business case for approaching potential container terminals operates and/or shipping lines to initiate dialogue on container handling at the port.

- Port profiling
- Proposed container berth
- Commercial structuring
 - Review the concession agreement of the port
 - Develop various business models
 - Revenue accruing to each party
 - Investment envisaged under different models
 - Return analysis, IRR, NPV
 - Determination of floor and ceiling for the client under various options for negotiation
 - Risk analysis
 - Prioritise the options and help in negotiations

PORT OF SINGAPORE AUTHORITY IS NOW A JOINT VENTURE PARTNER

29. Name of the assignment: Detailed project reports for Oil berth at Paradip port

Year: 2008-2010

Location: Odisha, India

Client: Paradip Port Trust (a major port in Orissa)

Main project features: The study included:

- Traffic projection exercise that included analysis of iron ore logistics including analysis of rail movement, shipping analysis in terms of vessel sizes and chartering rates, port tariffs etc.
- Determination of economic and financial feasibility based on the investment requirements and expected revenues from the development of a general cargo berth and an oil terminal
- Detailed forecast for revenue, capital, and operating cost based on industry analysis
- Determination of the best configuration for berths under the constraints of integration with the existing layout, financial optimization, and operational optimization
- Environment impact assessment of the abovementioned facilities

Position held: Project Director

Activities Performed: Supervised the preparation of the detailed project reports for the iron ore berth and oil berth for the Paradip Port Trust

30. Name of the assignment: Assessment of opportunities for expanding the port facilities at Vadinar under Kandla Port Trust

Year: 2010

Location:Kandla, India

Client: Kandla Port Trust

Main project features: The project involved two stages. In stage 1, a number of opportunities have been assessed to scale up the port facilities at Vadinar. In stage 2, after approval from KPT, a detailed techno-economic feasibility needed to be undertaken.

In stage 1, based on secondary sources of information and visual inspection, technical parameters of the suggested side had been assessed.

Also, a detailed study of hinterland and coastal industries has been undertaken to assess the consumption and production pattern of commodities in dry bulk, liquid bulk, crude, POL, container, project cargo, and break bulk.

From the above analysis, a detailed Traffic Assessment, in terms of overall volume and future growth for each commodity group, has been carried out. A decision matrix was prepared to shortlist the opportunities to develop at the suggested site.

Position held: Project Director

Activities Performed: Led the project for the Kandla Port Trust and reviewed the output

31. Name of the Assignment: Traffic study and broad feasibility study for Jodiya port in Gujarat under the Auspices of Gujarat Maritime Board (GMB)

Year: February 2010 – June 2010

Location: Gujarat, India

Client: Gujarat Maritime Board

Main Project Features:

Position held: Project Director & Team Leader

Activities Performed:

- Assessed coal volumes that could be attracted by undertaking a detailed study of hinterland industries
- Interviewed key stakeholders such as coal traders, industries, and port officials to understand the conditions (such as pricing and level of mechanization) to attract the cargo at Jodiya from competing port facilities
- Explained the pros and cons of direct berthing and lighter age option, based on traffic potential and technical parameters
- Recommended the way forward

32. Name of the assignment: Study mechanization of Magdalla Port in Hazira Gujarat GMB

Year: September 2009 – December 2009

Location: Gujarat, India

Client: Gujarat Maritime Board

Main project features:

- Examined the existing volume handled at the lighterage port of Magdalla
- Projected the traffic in the wake on hinterland activity and new developments such as award of ports at Hazira and Dahej to Adani, Essar, and Sterling.
- Based on traffic, undertook a detailed analysis of bottleneck in the chain involving Magdalla. The aspects included channel capacity in the context of tidal variations, level of mechanization, connectivity analysis, storage availability, and evacuation planning
- Suggested ways to remove bottlenecks in terms of change in evacuation plan, increased mechanisation, etc.

Position held: Project Director

Activities Performed: Recommended the way forward and supervised the study on the mechanisation of the Magdalla Port

33. Name of the assignment: Traffic study & financial feasibility for the mechanisation of Berth 5 at Kakinada deep water port

Year: June 2011- November 2011

Location: India

Client:Bothra Shipping Services Private Limited

Main project features:

Traffic Study

- Identified the primary, secondary, and tertiary hinterland for the steam/thermal coal and metallurgical/coking coal
- Conducted a hinterland study to map the cargo to hinterland clusters
- Assessed the existing and future hinterland connectivity through rail and road linkages
- Analyzed competition to assess the impact of existing and future expansion of port facilities that could compete for the cargo
- Forecasted cargo to and from the hinterland origin/destination points. This will provide the total potential traffic in the hinterland.
- Assessed the relative cost advantage of overall logistics cost for Kakinada vis-à-vis competing port facilities
- Based on the relative cost advantages and other factors such as ramp-up time and captive traffic, apportioned the traffic between competing port facilities
- Forecasted the coal traffic potential for the Kakinada Port covering the entire concession period

Financial Study

- Undertook the coal tariff analysis for the Kakinada Port
- Forecasted the capital expenditure, operational expenditure, and revenue
- Assessed the impact of tax benefits available to infrastructure ports.
- Developed the cash flow, balance sheet, and profitability statements
- Calculated the IRR, DSCR, and other relevant financial ratios and the project cost
- Undertook a sensitivity analysis to identify the risks
- Provided clarifications and explanation to lenders/financial institutions, if any, with respect to the project

Position held: Project Leader

Activities Performed:

- Prepared the framework of the study to be carried out by the team
- Participated in all the client interactions and presentations
- Reviewed the quality of output key
- Acted as an interface for the client

34. Name of the assignment: Traffic study for fertilizers & container berths and financial feasibility study for fertiliser berth at Kakinada deep water port

Year: August 2012- February 2012

Location: India

Client: Bothra Shipping Services

Main project features:

- Container Traffic Estimation
 - Evaluate the container traffic in India and analyse the growth trends over the past, growth drivers, vessel size, and key trade routes of the container traffic in India
 - Undertake primary interactions and analyse key cargo generators and movements for container traffic
 - Undertake an ICD/CFS-level analysis, specifically in the hinterland, and assess existing and prospective container traffic potential

- Assess the destinations and origin points that could potentially prefer east coast and KDWP for their trade
- Provide key success factors for the container terminal and success stories if any
- Fertilizer Traffic Estimates
 - Define hinterland for the port establishing key destinations that could be targeted
 - Evaluate the policy and regulations for the fertilizer traffic at ports and the impact that it could have on the traffic at the port
 - Assess cargo potential from primary and secondary hinterlands by evaluating various existing and upcoming user groups
 - Undertake limited primary interactions with select players in public, joint, and private sectors for evaluating key expansion plans and outlook
 - Use a top-down approach to assess the total traffic potential. Where applicable, the total logistics costs method will be employed to assess the potential for importing FRM.
 - Assess potential traffic that can be targeted by KDWP
- Traffic forecast for a period of next 18 years (concession period for KSPL) with additional scenarios to reflect 28 and 38 years.
- Undertake competitor analysis based on publicly available information as well as market intelligence
- Provide case on the ideal facilities for additional expansion at KDWP with reference to containers and fertilizer traffic
- Financial Analysis (Fertilizer Berth)
 - Estimate the block capital expenditure, operational expenditure, and revenue streams
 - Assess the impact of tax benefits available to infrastructure ports.
 - Develop the cash flow, balance sheet, and profitability statements
 - Calculate the IRR, DSCR, and other relevant financial ratios and the project cost
 - Undertake a sensitivity analysis to identify the risks

Position held: Project Leader

Activities Performed:

- Prepared the framework of the study to be carried out by the team
- Participated in all the client interactions and presentations
- Reviewed the quality of output key
- Acted as an interface for the client

35. Name of the assignment: Techno-economic feasibility study to develop Colachel as transshipment hub

Year: 2011

Location: Tamil Nadu, India

Client: Tamil Nadu Maritime Board, planning body in Tamil Nadu

Main project features: The salient features of the assignment are:

- Analysis of factors affecting the choice of ports as transshipment hub by shipping lines. This included a broad analysis of cargo potential, deviation from the shipping route and associated costs, regulatory framework such as cabotage laws, etc.
- Developing a strategy for technical and financial feasibility of developing a major container transshipment hub and bulk port competing with the regional leaders such as Singapore,

Colombo, Dubai, and Salalah

- Undertaking a traffic study for container traffic for the Port at Colachel
- Developing scenarios in the wake of development of Vallarpadam and Vizhinjam and competition from Colombo, Singapore, Salalah, etc.
- Advising the board on the extent of viability gap funding for the project
- Suggesting the PPP models for developing the port facilities

Position held: Project Director & Team Leader

Activities Performed: Interacted with TNMB on continuous basis, presented the findings of the study to senior bureaucrats of Tamil Nadu, and reviewed and managed the quality

36. Name of the assignment: Feasibility of bringing bigger vessels into Haldia Dock Complex carrying coking coal and assessment of trans-loading option for Steel Authority of India

Year: April 2010 to October 2010

Location: Haldia, India

Client: Steel Authority of India Limited (SAIL)

Main project features: The assignment included developing a shipping solution for SAIL to import coking coal for its plants:

- Identified a suitable site based on secondary sources of information and discussions with port officials
- Checked the feasibility of bringing cargo in a bigger vessel (Capesize/Panamax) by way of a mid-sea trans-loading operation at a suitable site to shift cargo to smaller vessels
- Developed other solutions such as:
 - Transshipment at the Dhamra Port
 - Direct unloading through the Dhamra Port to serve SAIL's plant via rail evacuation
 - Unloading through floating cranes at the Haldia anchorage
- Shipping cost analysis from load port (Indonesia, Australia) to the destination
- Charter rate forecast
- Cost-benefit analysis for each option: including savings in logistics cost, investment requirement, service-level issues, efficiencies of evacuation, etc.
- Calculated financial indicators like net present value (NPV) and internal rate of return (IRR)
- Recommended the best mode of unloading coking coal based on the above analysis

Position held: Project Director & Team Leader

Activities Performed: Supervised the entire study and presented the findings to the Chairman and Managing Director (CMD) and the board.

37. Name of the Assignment: Preparing an entry strategy for Fourcee in coastal shipping as part of its strategy to provide comprehensive logistics solutions for cargo needing specialized containers

Year: 2010

Location: India

Client: A private player

Main project features: By entering coastal shipping, the client wanted to extend the gamut of intermodal operations by road and rake the coastal shipping mode as well. The study encompassed:

- Analysis of macroeconomic drivers for liquid logistics in India across key cargo categories relevant to the client
- Establishment of future outlook for domestic production and consumption of the cargo categories for domestic distribution and logistics
- Analysis of cost efficiency of logistics business by capitalizing on economies provided by the shipping mode
- Analysis of competition
- Suggestion of phase-wise expansion in terms of routes and investment
- Preparation of a detailed financial model including profit & loss accounts, balance sheet, and cash flow statements
- Calculation of financial parameters such as IRR, NPV and debt-service coverage ratio (DSCR)
- Developing an outsourcing strategy for non-core operations
- Suggesting a suitable organisation structure with job descriptions and reporting linkages
- Strengthening competitive position by leveraging the ISO tank container inventory across logistics and shipping operations
- Developing a scalability and control model for future business growth while de-risking the business model from overdependence on individual transportation modes

Position held: Project Director

Activities performed: Supervised preparation of entry strategy for the private player in coastal shipping as part of its strategy to provide comprehensive logistics solutions for cargo needing specialized containers

38. Name of the assignment: Developing an entry strategy for commencing container shipping services for an international client

Year: 2009-2010

Location: Bangladesh

Client:

Main project features: The key elements of the study were:

- Identifying potential nodes for plying container vessels based on traffic volume and frequency of volume availability
- Developing a mathematical model to assess the viability of running services between the nodes
- Undertaking complex modelling for various cost elements and assessing the economic volumes that can justify vessels services from respective nodes
- Developing a business case by assessing the likelihood of minimum volume and its impact on the model
- Determining various financial indicators such as NPV, IRR and DSCR to assess the attractiveness from the perspective of various stakeholders

Position held: Project Director

Activities performed: Led the team for developing the entry strategy

39. Name of the assignment: Business plan (India Maritime Strategy) for developing a maritime city in India

Year: 2009

Location: India

Client: Zoom Developers

Main project features: The project involved developing facilities for marine tourism, ports and shipyards as a part of the proposed city. Salient project features were:

- Identifying an exhaustive set of locations, encompassing the entire Indian peninsular coastline, to identify three most suitable locations from the maritime city's perspective. The traffic study included a financial appraisal and assessment of technical suitability.
- Assessing opportunity and site for developing a shipyard within the maritime city
- Developing a strategy for a self-sustaining city
- Design of the city with employing synergies to reduce the capital cost of the various sub-projects

Position held: Team Leader

Activities Performed: Review of output quality, supervision and developing a key interface for the client

40. Name of the assignment: Logistics study for a major port-cum-logistics private player - Mundra Port and SEZ (MPSEZ)

Year: 2009

Location: India

Client: A major port-cum logistics private player

Main project features: The principal components of the study were:

- Logistics analysis in terms of cost, service and promotion of 20 commodities from origin to destination, with client's portfolio of port and its competition
- Collection of data on volumes handled, business channels, mechanization levels, inefficiencies, handling cost, pilferage and losses, etc.
- Analysis of customer preferences
- Value chain analysis to assess the possibility of reconfiguring, modifying, or removing some segments to serve end customers in an efficient and cost-effective manner
- Evolved a marketing strategy and highlighted investment opportunities

Position held: Project Director

Activities Performed: Prepared framework of the study to be carried out by a team of six people and participated in all the client interactions and presentations

41. Name of the assignment: Preparation of master plan, business plan and techno-economic feasibility study for Karaikal Port

Year: 2008

Location: India

Client: A private port player in South India

Main project features: The port has started operations recently. The principal outcomes of the project were:

- Detailed financial appraisal of first operating private port in south India
- Market projections (traffic study) for different types of cargo based on existing and future potential
- Phase-wise budgeting assessment in terms of civil, mechanical, electrical and other cost heads with the objective of enhancing value of the project
- Pricing strategy and tariff estimation
- Calculating operational expenses based on a detailed bottom-up approach
- Estimation of various financial indicators like DSCR, NPV (using DCF approach), and IRR for different sets of investors
- Preparing a financial model for project financing and assisting client in debt and equity syndication for the port project
- Preparation of master plan encompassing future expansion plans

Position held: Team Leader

Activities performed: Led team in preparing the master plan and business plan and undertaking techno-economic feasibility study

42. Name of the assignment: Pre-feasibility study for Porbandar and Jaffrabad ports on behalf of Chennai-based real estate and infrastructure developer

Year: 2008

Location: India

Client: AChennai-based real estate and infrastructure developer

Main project features: The scope of work included:

- Traffic study for both ports including study of thermal coal, coking coal, fertilizers, etc., that included assessment of the existing and future trends in the context of preferring the ports under study
- Site visits to gain a first-hand understanding of sites being studied
- Study of available technical reports such as admiralty data and hydrographic maps
- Capacity analysis of ports from a technical point of view
- Development of conceptual layout for both the ports
- Block cost estimation for projected capital expenditures, fixed and variable costs and financing costs
- Preparation of proforma financial statements, such as balance sheets and P&L
- Calculation of various financial indicators such as NPV, IRR, and DSCR
- Preparation of a comparison matrix to examine the feasibility and preference between the Porbandar and Jaffrabad ports
- Recommending the best option to the client

Position held: Team Leader

Activities Performed: Was involved in the entire course of study, developed the framework, reviewed the progress, and acted as the key interface for the client

43. Name of the Project: Traffic study for Fertiliser and container berths and financial feasibility study for fertiliser berth at Kakinada Deep Water Port

Year: August 2011 to February 2012

Location: East Godavari, Andhra Pradesh, India

Client: Bothra Shipping Services

Main Project Features: Bothra Shipping Services (BSS) has entered into a memorandum of understanding (MoU) with Kakinada Sea Ports Limited (KSPL) for mechanisation and exclusivity for coal handling at berth No. 5 and management of the storage and port evacuation facilities. BSS also intends to extend its activities to develop facilities at two upcoming berths to meet growing demand for bulk and container facilities in the hinterland. To validate the preliminary understanding, BSS intends to undertake a formalized traffic study to understand:

- a. The potential for container terminal in the region
- b. The growth potential for fertilisers

BSS also plans to study the financial feasibility for a fertiliser berth.

Position held: Project Leader

Activities performed: The scope of services included:

- Estimating container traffic
 - Evaluated container traffic in India, past growth trends, growth drivers, vessel size analysis, and key trade routes of container traffic in India
 - Undertook primary interactions and analysed key cargo generators and movements for container traffic
 - Undertook a ICD/CFS-level analysis specifically in the hinterland and assessed container traffic potential
 - Assessed destination and origin points that could potentially prefer east coast and Kakinada Deep Water Port (KDWP) for their trade
 - Provided key success factors for the container terminal and success stories if any
- Fertilizer traffic estimates
 - Defined the port hinterland, establishing key target destinations
 - Evaluated policy and regulations for fertilizer traffic at ports and presented their impact on traffic at the port.
 - Assessed cargo potential from primary and secondary hinterlands by evaluating existing and upcoming user groups.
 - Undertook limited primary interactions with select players in the public, joint and private sectors for evaluating key expansion plans and outlook
 - Used top-down approach to assess the total traffic potential. Where applicable, the total logistics cost method was employed to assess the potential for importing FRM.
 - Assessed potential traffic that can be targeted by KDWP
- Traffic forecast for next 18 years (concession period for KSPL) with additional scenarios to reflect 28 and 38 years forecasts
- Undertook competitive analysis based on publicly available information and market intelligence
- Provided a case on ideal facilities for expansion at the KDWP with reference to containers and fertiliser traffic

- Financial analysis (fertiliser berth)
 - Estimate block capital expenditure, operational expenditure and revenue streams
 - Assess impact of tax benefits available to ports
 - Develop cash flow, balance sheet and profitability statements
 - Calculate the IRR, DSCR and other relevant financial ratios and estimate project cost
 - Undertake sensitivity analysis to identify the risks

B. PORTS & SHIPPING (PPP & TRANSACTION ADVISORY)

44. Name of the assignment: Transaction advisory for mechanisation of iron ore handling berth WQ1 in the inner harbour at Visakhapatnam port

Year: June 2010 – June 2014

Location: Visakhapatnam, India

Client: Visakhapatnam Port Trust

Main Project Features: The Visakhapatnam Port Trust (VPT) had invited applications from private players for mechanizing the WQ1 berth for handling iron ore on PPP basis. The engagement involved assisting VPT in conducting the bid process and selecting a private player for the same.

Position Held: Project Leader

Activities Performed: Supervising and leading the team for the assignment, which includes the following tasks:

- Preparing documents for obtaining TAMP approval for upfront tariff
- Evaluating applications received for qualification and providing assistance in shortlisting parties
- Preparing and finalizing tender documents, namely, Request for Proposal (RFP), Draft Concession Agreement (DCA), and PPPAC Memo
- Evaluating the financial proposals received and assisting in finalizing the successful bidder
- Assisting VPT in signing the concession agreement
- Assisting in the selection and appointment of an independent engineer for the project

45. Name of the assignment: Pre-bid advisory for berth No. 14 at Kandla port

Year: June 2012 - July 2012

Location: Gujarat, India

Client: ACT Infraports Limited

Main Project Features: The Kandla Port Trust (KPT) had invited bids from suitable private sector entities to develop berth number 14 as multipurpose cargo berth. Towards this purpose, CRIS had advised one of the shortlisted bidders, ACT Infraports (ACT), in the preparation of a responsive, optimal, and competitive bid for the project.

Position Held: Project Leader

Activities Performed: The activities performed under this assignment include the following:

- Analysis of available documents and bid assistance

- Broad analysis of the cargo patterns
- Tariff and revenue analysis
- Phase-wise block cost estimate (especially for equipment) of capital and operational expenditure
- Assessment of the likely financing structure and expected financing costs
- Financial modelling
- We advised the client not to bid for the project

46. Name of the assignment: Transaction advisory services for installation of mechanized fertiliser handling facilities at East Quay-7 (EQ-7) in the inner harbour of Visakhapatnam port on design, build, finance, operate, and transfer (DBFOT) basis

Year: August 2010 - November 2013

Location: Visakhapatnam, India

Client: Visakhapatnam Port Trust

Main Project Features: The Visakhapatnam Port Trust (VPT) intends to undertake the installation of mechanised fertiliser handling facilities on East Quay-7 in the inner harbour of the Visakhapatnam port through public-private partnership on design, build, finance, operate, and transfer (DBFOT) basis for a concession period of 30 years.

Position Held: Project Leader

Activities Performed: Supervising and leading the team for the assignment, which involves activities such as:

- Evaluating applications received for qualification and providing assistance in shortlisting parties
- Preparing and finalising tender documents, namely, Request for Proposal (RFP) and Draft Concession Agreement (DCA)
- Evaluating the financial proposals received and assisting in finalizing the successful bidder
- Assisting VPT in signing the concession agreement
- Assisting in the selection and appointment of an independent engineer for the project

47. Name of the Assignment: Transaction Advisory for Chennai Port Trust

Year: 2008-2011

Location: Chennai, India

Client: Chennai Port Trust

Main project features: The key work undertaken was:

- Financial feasibility study for the proposed mega terminal at the Chennai Port
- Managing the consortium as the lead partner
- Preparation of bid documents
- Preparation of documents for meeting regulatory requirements
- Evaluation of bidders and preparation of RFQ report
- Assisting the Chennai Port Trust in representation to regulatory and statutory bodies such as PPPAC and TAMP

Positions held: Project Director & Team Leader

Activities Performed: Prepared the framework of the study to be carried out by the team, participated in all the client interactions and presentations, supervision, reviewed the output quality, and acted as the key interface for the client

48. Name of the assignment: Advise on bidding strategy for bidding for NCB-2 through PPP model at Tuticorin port

Year: 2010

Location: Tuticorin, India

Client: A private sector player

Main project features: The key aspects of the assignment include:

- Traffic projection for coal
- Analysis of RFP document to understand the key technical and commercial parameters and its implication on volume and thus bidding strategy
- Broad financial analysis to establish the bidding scale
- Scenario development based on business intelligence on competition
- Suggesting bidding strategy
- Preparing bid documents on behalf of the client

Position held: Team Leader & Project Director

Activities Performed: Acted as the key interface with the client and suggested the best bidding strategy to the client

49. Name of the assignment: Bid assistance for port at Nagapattinam, Tamil Nadu, through PPP route

Year: 2010

Location: Tamil Nadu

Client: A private port developer and operator

Main project features: The main areas covered were:

- Traffic analysis for dry bulk including captive and third party cargo
- Developed number of scenarios based on indicative financial numbers
- Suggested the best and winning bidding strategy

Position held: Project Director

Activities Performed: Involved with client to assist them in best bidding strategy and put together the documents for bidding.

50. Name of the assignment: Assistance to a major infrastructure player in bidding for JNPT's fourth container terminal through the DBFOT route

Year: 2010

Location: India

Client: A major infrastructure player

Main project features: The assignment covered:

- Studied the value chain for container logistics in India in a detailed manner that included cost, quality, and service assessment of various elements in the chain including analysis of shipping, port, inland logistics and inland infrastructure such as container freight stations (CFS) and inland container depots (ICD).
- Study on container traffic at the proposed facility
- Competition analysis for mapping current and proposed plans for augmenting container handling system at competing ports such as Mundra, Kandla, ICT (MBpt), Pipavav and upcoming ports at Dahej, Dighi, and Hazira
- Analysis of data collected through primary surveys at the ports along the western coast to understand the hinterland and commodity break-up
- Secondary sources of data had been utilised to assess the traffic movement pattern from ICD, CFS and direct traffic
- Container volume was further split to estimate the proportion of 20/40 foot, empty/laden, export/import, etc
- Analysis of the average dwell time and its implication on storage both inside the port and CFS
- Detailed connectivity analysis for sufficiency of rail and road infrastructure to handle the proposed volume
- Study of constraints to examine the other bottlenecks related to marine and land infrastructure

Position held: Project Director & Team Leader

Activities performed: Provided traffic estimates through a top-down and bottom-up approach and validated the same through various sources of information, supervised the team, and reviewed the output quality

C. PORTS & SHIPPING (TRAFFIC & OTHERS)

51. Name of the Assignment: Due diligence for various port assets

Year: February 2015-November 2015

Location: Gujarat

Client: Essar Ports Limited

Main Project Features: Essar Ports is one of the leading private sector port developers and operators in India for handling liquid, dry bulk, break bulk and general cargo. It has an existing aggregate capacity of 104 MTPA across two facilities located at Vadinar and Hazira in the state of Gujarat on the west coast of India and one dry bulk export terminal at Paradip in the state of Orissa on the east coast of India and an iron ore facility in Vizag Port.

The primary objective of this exercise is to undertake a market due-diligence of the various assets owned by Essar Ports Limited under various concessions across India and to conduct a tariff benchmarking for competing facilities for key commodities.

Position Held: Project Leader

Activities Performed: The scope of work included:

- Market valuation

- Tariff benchmarking
 - Spot tariff
 - Long-term contracts

The tariff benchmarking was conducted for the key competing facilities and covers the following ports/terminals:

- For assets in Gujarat (Hazira, Salaya, Vadinar): Mundra, Kandla, Pipavav, Dahej, Adani Hazira, Mumbai, Bedi and Navlakhi. For Bedi and Navlakhi, the assessment for coal tariff will be done considering lighterage operations and will include the entire charges applicable (lighterage, movement by barges, unloading at jetty etc.)
- For assets in Paradip: Kolkata, Dhamra, Visakhapatnam and Gangavaram
- For assets in Visakhapatnam: Gangavaram, Kakinada, Krishnapatnam

52. Name of the Assignment: Assessment of coastal cargo in Gujarat

Year: November 2014 – July 2015

Location: Gujarat, India

Client: Gujarat Infrastructure Development Board

Main Project Features:

Gujarat Maritime Board (GMB) intends to assess the potential of coastal cargo movement in the state of Gujarat. The Authority, through a tendering process taken up by Gujarat Infrastructure Development Corporation (GIDB), has selected CRISIL Risk & Infrastructure Solutions (CRIS) to prepare a report for assessment of coastal cargo potential in Gujarat. The scope of work entails assessment of interstate movement of commodities; understanding the coastal cargo movement in India and Gujarat; assessing possible modal shifts and therefore potential of future coastal movement of cargo in Gujarat, apart from evaluating other factors such as regulations impacting coastal cargo and providing a list of viable routes/commodities.

Position Held: Port sector specialist

Activities Performed: Supervision, presentations to client

53. Name of the assignment: Consolidated study on ports

Year: November 2013 – April 2014

Location: Gujarat, India

Client: Essar Ports Limited

Main Project Features:

CRISIL Infrastructure Advisory assisted Essar Ports Limited in three independent studies:

- Preparing the traffic study report for coal at the proposed bulk terminal at Salaya.
- Preparing a traffic study report for Iron ore at Visakhapatnam port.
- Developing a concept presentation on a mega port at Salaya.

Position Held: Team Leader

Activities Performed:

- Module 1: Estimation of traffic potential for coal and bauxite at Salaya port.
- Module 2: Module 2 refers to the preparation of a presentation on Salaya port

- Module 3: Traffic study for estimating the iron ore traffic potential (coastal and exports) from the proposed three iron ore berths at Visakhapatnam port

54. Name of the assignment: Traffic & market Assessment for JNPT's fourth container terminal

Year: October 2013 – May 2014

Location: Maharashtra, India

Client: Confidential

Main Project Features: JNPT, the largest container port in the country is planning to increase its capacity by developing the fourth container terminal which would add capacity of 4.8 MTEUs per annum, to be developed in two phases. The Trust has invited applications for development of the project on a PPP basis and the client is pursuing the same. CRISIL Infrastructure Advisory was appointed to undertake a market assessment study for the new terminal. The assignment aims at analysing the traffic potential for JNPT for the 30-year concession period.

Position Held: Team Member

Activities Performed:

- Defined and studied the hinterland in terms of primary, secondary and tertiary hinterland. Based on the above analysis, as well as macro-economic factors, analysed hinterland cargo potential for next 30 years
- Provided overview of competition from other existing container handling facilities in west coast of India and a forecast for future development of additional container handling facilities on west coast
- Determined the share of traffic to other ports/ terminals and carried out competition landscape
- Assessed perception of leading shipping lines and potential users on the proposed facilities at JNPT's fourth container terminal
- Assessed developments in hinterland over medium and long-term and analysed critical success factors for the container terminal

55. Name of the assignment: Concept presentation on a mega port at Hazira: The trade gateway of India

Year: August 2013 – November 2013

Location: India

Client: Essar Bulk Terminal Limited

Main Project Features: Essar Ports Limited is keen to develop a concept presentation on the Hazira Mega Port to position itself as the trade gateway of India.

Position Held: Project Leader

Activities Performed: The scope of work encompasses the following:

- Assessment of benefits by creation of port markets and development of case for port market.
- A broad road-map of having maritime cluster with port market as nucleus of development
- Quick assessment of similar concepts abroad such as Singapore or Dubai

- Assessment of Hazira as an ideal location for the development of Mega Port
- Broad assessment of impact of the maritime cluster in terms of industries likely to come
- Broad quantification of maritime cluster impact in terms of multiplier effect for employment, output and investment
- Broad assessment of impact on logistics cost for bulk and container traffic and brief case for increased contribution in trade competitiveness for the country
- Showcase the payback to the government (both state and central) in terms of taxes due to cluster development

56. Name of the assignment: Consultancy services for approval of up-front tariff fixation by TAMP for IKBL Barge Jetty for third party cargo handling

Year: July 2013-March 2016

Location: India

Client: Indian Farmers Fertilisers Co-operative Limited (IFFCO)

Main Project Features: M/s IFFCO Kisan Bazar and Logistics Limited (IKBL) is setting up Captive Barge Jetty of 120m*20m size with 2 MMTPA capacity. The jetty shall handle imported solid raw materials and finished fertiliser products. It will help reduce waiting time at ports for ships bringing IFFCO cargo, thereby enabling receipt of material on KPT berth which is usually congested.

A Model Concession Agreement was signed on February 17, 2011 between IKBL & KPT. As per Article of Concession agreement, "Levy and recovery of the tariff", it is mentioned that tariff is not levied for own use. In case the concessionaire handles third party cargo as directed by concessioning authority, it shall be entitled to levy and recover tariff from the owners/consignees of cargo and vessel owners/agents using the project facilities and services as per the scale of rates notified time to time. It is also stated in the article that the cargo handling charges shall be as approved by TAMP. Hence, application has to be submitted to TAMP for approval of tariff for handling third party cargo charges. TAMP has laid down the guidelines for upfront PPP (Public Private Partnership) project tariff setting at major port trusts.

Position Held: Project Leader

Activities Performed: The scope of work included the following modules:

- Preparation of TAMP application for fixing up-front tariff for third party cargo handling for IKBL barge jetty as per TAMP guidelines for PPP projects, 2008
- Arriving at the rate for which TAMP approval is sought and the same to be explained and got approval by IKBL before submission to TAMP
- To follow any new guidelines given by TAMP during course of obtaining approval and ensuring its compliance in application
- To reply and submit to TAMP all the queries raised by it
- To obtain an approval for third party cargo handling at the IKBL barge jetty

57. Name of the assignment: Commercial and market due diligence of a private freight terminal

Year: August 2013 – January 2014

Location: India

Client: Confidential

Main Project Features:

Position Held: Project Leader

Activities Performed: The scope of work includes the following tasks:

- Initial data review
 - Data collation and interactions
 - Sector overview
- Market, technical and commercial review
 - Market diligence
 - Commercial diligence
 - Technical diligence
- Final due-diligence report

58. Name of the assignment: Due diligence for a port opportunity

Year: March 2013 – September 2013

Location: India

Client: Haskoning India Pvt. Ltd.

Main Project Features: Commercial due diligence for a port in India

Position Held: Project Leader

Activities Performed: The scope of work includes the following tasks:

- Estimation of current demand (market) and expected growth, which included traffic study and market assessment
- Review of the company's business plan (management case) and expected strategic identification of potential areas of threat from the competitors
- Estimation of current market share and projection of expected market share based on company's plans and expected competitive pressures by reviewing:
 - Growth plans, investment plans, strategic goals, and
 - Review of traffic studies, and other market studies
- Review of the main underlying assumptions for the cargo throughput, such as progress of developments in the hinterland (e.g. rail capacity, industrial zone developments) or specific projects of material importance to the realisation of the cargo throughput
- Assessment of alliances with major partners, such as international major shipping lines and CONCOR.

59. Name of the assignment: Traffic forecast for development of multipurpose port in Kakinada extension study

Year: December 2012 – April 2013

Location: India

Client: GMR Infrastructure Limited

Main Project Features: Andhra Pradesh is one of the most progressive states in India with a dynamic economy. An investor-friendly climate prevails in the state, making it one of the most attractive destinations for investments. Taking advantage of these aspects and as a part of the development programme, Kakinada SEZ intended to develop a multi-cargo, all-weather gateway port near Kakinada. Kakinada SEZ is a port-based development along the Andhra Pradesh coast and has been promoted by Kakinada SEZ Pvt. Ltd. The company proposed to develop an all-weather, deep-water, multi-purpose, world-class port in the vicinity of Kakinada SEZ (KSEZ) in Kona village, Thondangimandal, East Godavari district, Andhra Pradesh state, India.

Position Held: Project Leader

Activities Performed: The scope of work for the study included the following modules:

- To assess the likely traffic potential of the proposed multi-purpose port in Kakinada SEZ, establishing a baseline traffic forecast and potential traffic scenarios over a long-term planning horizon
- Identification of key trade and economic development trends that are shaping India, Andhra Pradesh and the Kakinada port catchment area/ hinterland
- Port development opportunities emerging from the medium and longer-term scenarios/trends
- Identification of key success factors to leverage Kakinada's SEZ position and win new development and revenue sources

60. Name of the assignment: Study for assessing overall traffic potential at Kakinada deep water port

Year: December 2012 – November 2013

Location: India

Client: Kakinada Seaports Private Limited

Main Project Features: Kakinada Deep Water Port (KDWP) is located in the East Godavari district of Andhra Pradesh and is situated between the major ports of Visakhapatnam (about 170 kms North) and Chennai (about 650 kms South). The port was constructed with a quay length of 610 meters by Government of Andhra Pradesh and it was commissioned in November 1997. The privatisation process was based on the international bidding in (B)OMST (Build Operate Maintain Share and Transfer) format which was awarded to Kakinada Seaports Private Limited (KSPL). The project was commissioned in 1997 and started operations from April 1999.

Position Held: Project Leader

Activities Performed: The scope of work as per the document issued shall include the following modules:

- Identification of key commodities and traffic forecasting
- Analysis of port connectivity and needed infrastructure
- Vessel analysis and impact of dredging
- Impact assessment of setting up of LNG terminal on creating industrial demand from hinterland
- Traffic assessment and overall market overview for OSVs in KG Basin

61. Name of the assignment: Assisting in exploring opportunities for JSW Jaigarh Port with DHL

Year: September 2012

Location: India

Client: JSW Infrastructure

Main Project Features: The scope of work included:

- Discussing specific requirements of the identified player and exploring the direction for discussions and agreement
- Arranging and participating in the meetings between the client and the identified partner
- Assisting the client in signing a memorandum of understanding (MoU) establishing the interest between parties to take the proposal to the next level
- Assisting JSW in exploring various options of appropriate commercial project structuring
- Assisting the signing of contract between the client and the identified partner and in finalising the deal
- Assisting JSW in the preparation of the business plan

Position Held: Project Leader

Activities Performed: Supervision, review of quality of output, key interface for the client

62. Name of the assignment: Hand-holding assistance for review of port in Andhra Pradesh

Year: November 2011- December 2013

Location: India

Client: Flowers and Petals.Com

Main Project Features: INCAP, an arm of I&I department, mandated CRIS to undertake a detailed review of various agreements with Gangavaram Port Limited and other documents as from the perspective of all stakeholders with a view to reconcile any past deviations in the execution of the assignments and to establish a well-defined framework to monitor the same going forward.

Position Held: Project Leader

Activities Performed: The scope of work encompasses the following:

- To undertake a detailed review of various agreements and other documents
- To formulate a monitoring framework for the GoAP
- To provide a hand-holding assistance for initial years to closely assist GoAP in implementing the framework and being part of GoAP representation teams for all interface with SPV

63. Name of the assignment: CII-CRISIL collaborative study on port-led development in the western states

Year: 2011

Location: India

Client: Gujarat Maritime Board

Main Project Features: The study involved analysis of:

- Port scenario in western India - Major/minor ports
- Growth trends in MH/Gujarat/Goa ports (capacity/cargo handled - 10 yearly trends/future growth projections) vis a vis all Indian ports
- Hinterland analysis - cargo belts - development of ancillary industries.
- Port connectivity issues - rail road linkages - current initiatives (Govt/Pvt) to implement port-rail projects (establishment of SPVs)
- Developments in the port services and logistics sector
- Port policies of respective state governments - their efficacy in spurring port-led development - support to private initiatives in the sector

Position held: Team Leader

Activities Performed: Supervision, review of quality of output, key interface for the client

64. Name of the assignment: Traffic forecast for development of multipurpose port in Kakinada

Year: April 2011 –July 2012

Location: India

Client: AECOM India (Pvt.) Limited

Main project features: During the study, the team would:

- Map the hinterland of the port that could potentially generate traffic
- Determine the market potential for key cargo commodities including coal, iron ore, alumina, limestone, cement, food grains, POL, container, and others that can be handled at the port
- Undertake a hinterland analysis of the port
- Undertake a hinterland traffic potential and vessel analysis
- Undertake a competitive analysis of the major and non-major ports in proximity
- Evaluate the total logistics cost to evaluate the traffic that can be targeted by the port
- Assess the traffic potential that could be created and be shifted to the port
- Project traffic for the next 25-30 years
- Undertake a sensitivity analysis of the projected traffic
- Identify and estimate project revenue streams
- Estimate project capex including infrastructure development costs (both marine and non-marine)
- Estimate port operating expenses and arrive at margin/profit estimates
- Project cash flows considering the project execution period, phasing of investments, revenues, and operating expense projections, funding mix, etc.
- Determine the financial feasibility (MS Excel model with 30-year horizon) – projection of IRR, NPV, pay-back, department service coverage etc. Along with scenario analysis based on the traffic projections and the sensitivity to key variables
- Output/insights from the financial feasibility analysis should be used to refine/optimize the overall port master planning.

Position held: Project Leader

Activities Performed:

- Prepared the framework of the study to be carried out by the team
- Participated in all the client interactions and presentations

- Reviewed the quality of output key
- Acted as an interface for the client

65. Name of the assignment: Revalidation of traffic potential for Subarnarekha port

Year: April 2012- March 2013

Location: Orissa, India

Client: Subarnarekha Port Private Limited

Main Project Features: In 2011, CRIS assisted the upcoming port at Subarnarekha in Orissa by undertaking a traffic study. Again, in 2012, the Subarnarekha port required the previous study to be re-visited to reflect recent changes and key developments since the previous study.

The broad objective of the study is to assess the changes in the external environment, identify the key developments in the port's hinterland, determine the change in competitive landscape, and assess the impact of these changes on the upcoming port's traffic potential in the short, medium, and long-term.

Position Held: Project Leader

Activities Performed: The activities performed under this assignment include the following:

- Scanning the external environment for
 - Regulatory/policy changes such as new policy announcements, restrictions in trade, or changes in regulations
 - Developments on the industry and infrastructure-side such as setting up of new industries in the hinterland and new connectivity linkages formed on road/rail
 - Changes in the competitive landscape such as new greenfield ports or brownfield expansions at the existing competing ports
- Identifying the key developments that are likely to impact the potential for the port and assessing the likely impact on the traffic potential of the upcoming port
- Providing recommendations on the broad strategy on how the Subarnarekha port can best position itself to benefit from the changes in its external settings

Taking into account the change and developments listed above, a reassessment of traffic potential for the upcoming port was undertaken. This exercise involved:

- Assessment of the hinterland traffic potential for the proposal port
- Competitive assessment of adjacent ports
- Updating of tariff and various charges at the ports in the hinterland
- Origin-destination analysis for the hinterland traffic
- Total logistic cost analysis for all the ports in the hinterland
- Traffic projections for the Subarnarekha Port

66. Name of the assignment: Calculation of ROCE for 2012-2013 for TAMP

Year: April 2012 - June 2012

Location: India

Client: Tariff Authority for Major Ports

Main project features:

Return on capital employed or ROCE is an important parameter that appears in TAMP guidelines. It acts as an important input towards setting tariffs at all major ports. TAMP had appointed CRISIL Infrastructure Advisory to reassess the ROCE.

ROCE was reassessed/calculated using the standard formula for the calculation of cost of capital, which included calculation of cost of debt and cost of equity separately.

Position Held: Project Director

Activities Performed: Supervision, review of quality of output, key interface for the client

67. Name of the assignment: Traffic forecast for Subarnarekha port

Year: February 2011 - May 2011

Location: India

Client: Subarnarekha Ports Private Limited

Main project features: Subarnarekha Port is proposed to be developed at the mouth of the Subarnarekha River in Balasore District, Odisha. The port is well-placed to cater to various types of cargo, especially bulk cargo, because of its location. Therefore, it could emerge as a potential port player catering to thermal and coking coal, iron ore, limestone, and container traffic.

Given the strategic location of the port, the client wanted a study to be conducted of the traffic that could be attracted to the port today and in the future, with the development of the hinterland. Also, the traffic was to be analysed in terms of the proposed port's competitiveness vis-a-vis other port facilities in the region.

Accordingly, the client approached CRISIL Infrastructure Advisory to provide assistance in traffic assessment at the Subarnarekha Port and provide a forecast for the same.

Position held: Project Leader

Activities Performed: The scope of work for this assignment included:

- Market potential of bulk and containerized commodities that can be handled at Subarnarekha Port
- Hinterland analysis of the port
- Hinterland traffic potential and vessel analysis
- Competitive analysis of the major and non-major ports in proximity
- Integrated logistics cost analysis of the port
- Assessment of the traffic potential that could be created and be shifted to the Subarnarekha Port, and future projections for the same
- Sensitivity analysis of the projected traffic

68. Name of the assignment: Assistance in tariff assessment of Subarnarekha port

Year: February 2011-May 2011

Location: India

Client: Subarnarekha Port Private Limited

Main project features: The client wanted a study to be conducted for estimation of the port tariff. Accordingly, the Client has approached CRISIL Risk and Infrastructure Solutions Limited (CRIS) to provide assistance in tariff assessment at Subarnarekha Port. CRIS is pleased to submit its proposal in this regard.

Hinterland Identification

- Identification of the existing and upcoming industry clusters and other cargo centres consuming/producing dry bulk cargo in the hinterland and dependent on the port
- Study of the existing rail and road network and upcoming rail and road projects
- Origin-destination analysis of all the cargo that can potentially come to proposed port
- Including interactions with select stakeholders
- Based on all of the above, definition of the hinterland (primary/secondary/tertiary) for the proposed port.

Competitive Assessment of Adjacent Ports

- Benchmarking of adjacent ports' (the nearest four major and non-major/private) infrastructure and proposed port's infrastructure
- Competitive analysis of adjacent ports in terms of:
 - Past trends in cargo profile, growth rates, and cargo movements
 - Existing cargo-handling facilities and proposed expansion plans
 - Road/rail connectivity to the cargo centres
 - Capacity analysis (which will be a function of channel capacity, berth capacity, equipment capacity, storage facilities, and evacuation)
 - Summary assessment of advantages/shortcomings of competing ports with the proposed port

Tariff Assessment for Proposed Port

- Calculation of voyage cost from the competing ports in India to the major international ports
- Evaluation of vessel and cargo related charges at the competing ports
- Analysis of inland transportation cost from the competing ports to the hinterland
- Rationalisation of the tariff at Subarnarekha by taking into account the total integrated logistics cost

Position held: Project Leader

Activities Performed:

- Prepared the framework of the study to be carried out by the team
- Participated in all the client interactions and presentations
- Reviewed the quality of output key
- Acted as the interface for the client

69. Name of the assignment: Proposal for contract vetting for BSS

Year: September 2011 - December 2011

Location: India

Client: Bothra Shipping Services

Main project features:

- Review the four contracts for:

- Commercial soundness
- Legal soundness
- Appropriateness of tax planning
- Ensure that the contracts meet the above criteria, make suggestions and necessary corrections
- Post-review, suggest alternative tax planning possibilities, if available
- Discuss with BSS various aspects of the agreements and address any concerns accordingly
- Prepare and submit a note on all such corrections and suggestions
- Provide clean contracts incorporating such comments acceptable to BSS

Position held: Project Leader

Activities Performed:

- Participated in all the client interactions and presentations
- Reviewed the quality of output key
- Acted as an interface for the client

70. Name of the assignment: Quick assessment of port traffic and shipyard facility at site near Vijaydurg Port in Maharashtra on behalf of a private infrastructure developer

Year: 2010

Location: Maharashtra, India

Client: A private infrastructure developer

Main project features:

- Analysis included studying constraints such as last-mile connectivity, availability of back-up land and marine parameters
- Broad assessment of market potential, especially for defence shipbuilding and ship repairs. Suggested the possible size and type including commercial ships that could be handled at the site.
- Broad assessment of traffic that the Vijaydurg Port could attract
- Examined commodities in dry bulk, liquid bulk, and break bulk categories
- Preparing a business case for investment in the proposed port

Position held: Project Director & Team Leader

Activities performed: Supervision, review of quality of output, key interface for the client

71. Name of the assignment: Traffic study for berth no. 7 at Mormugao Port to be developed (now awarded to Adani) via PPP route

Year: 2009

Location: Mormugao, India

Client: A big private port player

Main project features: The study covered:

- Analysis of coal logistics from the hinterland of Mormugao
- Interaction with more than 20 players including stevedores, traders and shippers

- Developing a quantifiable and probabilistic model to establish the base case for traffic that could be attracted
- Projection of cargo for entire concession period
- Sensitivity analysis to assess the variation in traffic with the changes in assumption

Position held: Project Director

Activities Performed: Led the team during the traffic study and ensured quality of deliverables

72. Name of the assignment: Business plan (India Maritime Strategy) for developing a maritime city in India

Year: 2009

Location: India

Client: A large real estate player

Main project features: The project involved developing facilities for marine tourism, ports and shipyards as a part of the proposed city. Salient project features were:

- Identifying an exhaustive set of locations, encompassing the entire Indian peninsular coastline, to identify three most suitable locations from the maritime city's perspective. The traffic study included a financial appraisal and assessment of technical suitability.
- Assessing opportunity and site for developing a shipyard within the maritime city
- Developing a strategy for a self-sustaining city
- Design of the city with employing synergies to reduce the capital cost of the various sub-projects

Position held: Team Leader

Activities Performed: Review of output quality, supervision and developing a key interface for the client

D. SHIPYARD

73. Name of the assignment: Entry strategy and technical feasibility study for a shipyard at Mugaiyur, Tamil Nadu

Year: 2010

Location: Tamil Nadu, India

Client: Shipyard

Main project features:

- Understanding technical features of the site in terms of marine, land, and connectivity parameters
- Analyzed advantages the site could offer in terms of logistics advantages and proximity to ancillary units
- Undertook a market study for shipbuilding and ship repair, as suited to the shipyard's technical feature
- Prepare a phase-wise layout for the shipyard, considering the operational efficiencies and return on investment
- Prepare a bankable report including financial assessment

- Prepare financial statements such as balance sheets, P&L statements, cash flow statements, and debt repayment schedule
 - Calculate ratios such as the NPV, IRR, and DSCR

Position held: Project Director

Activities Performed: Supervision, review of quality of output, key interface for the client

74. Name of the assignment: Study on development of ship repair and ship building strategy for EGYPT

Year: 2009 to 2010

Location: Egypt

Client: Government of Egypt (GOE)

Main project features: Managed the study for the client with the overall objective of suggesting a policy framework to facilitate development of the ship repair and shipbuilding sector in Egypt and suggesting measures to increase international competitiveness. The specific objectives included the following:

- Global review of the sector
- Assessed efficiency of shipyards in Egypt and performed a competitive analysis
- Identifying needed improvements
- Formulating a development strategy for the industry
- Preparing an action plan for the ship repair and ship building development strategy

Position held: Project Director & Team Leader

Activities Performed: Supervision, review of quality of output, key interface for the client

75. Name of the assignment: Bid advisory and submission for developing ship repair facilities at Kandla and a shipbuilding yard at Tuna

Year: 2009

Location: Kandla, India

Client: A private sector client

Main project features:

- Preparing a project feasibility report that encompassed:
 - Studying the ship repair and shipbuilding market in India
 - Technical study of the proposed facility
 - Financial modelling and return analysis
 - Suggestion of various development schemes
- Bid preparation and bidding strategy
- Preparing the presentation

Position held: Project Director

Activities Performed: Supervision, review of output quality, key interface for the client

76. Name of the Assignment: Techno-economic feasibility of a private-sector shipyards in Vengani, Gujarat

Year: 2009

Location: Gujarat, India

Client: A private-sector shipyard

Main project features:

- Market assessment of ship repair units
- Financial return analysis
- Detailed technical layout for proposed shipyard after considering efficiencies in material movement
- Suggested the level of mechanization

Position held: Team Leader

Activities Performed: Review of output quality, supervision, key interface for the client

77. Name of the assignment: Finding a suitable site for developing shipbuilding and ship repair facility in West Bengal

Year: 2009

Location: West Bengal, India

Client: Private-sector logistics player

Main project features: The study analysed:

- Technical parameters of various sites, based on secondary sources of information such as wave and current data, topographical features, and marine features
- Market for ships that could be built and/or repaired
- Decision matrix to finalise site for development, based on advantages/disadvantages and their weightages for building a ship-repair/shipbuilding unit
- Suggested two sites for development

Position held: Project Director & Team Leader

Activities Performed: Supervision, review of output quality, key interface for client

78. Name of the assignment: Advisory paper on subsidies to shipyards in India

Year: 2007

Location: India

Client: Shipyard Association of India

Main project features: Prepared an advisory paper on subsidies to shipyards in India; the paper is considered to be a landmark in the shipbuilding community. The study included:

- Assessed economic impact of shipbuilding on economy
- Benchmarking with other leading shipbuilding nations in terms of government support extended to the industry and quantification of disadvantages faced by the Indian shipyard industry owing to direct and indirect support provided by China and Korea, two major shipbuilding countries

- Developed a multiplier model, which involved creative thinking and co-ordination with various international offices of KPMG to glean the information

Position Held: Senior Consultant and module leader

Activities Performed:

- Involved in all the client interactions and presentations
- Reviewed quality of output
- Acted as an interface for the client

E. ROADS

79. Name of the Assignment: Consultancy Services for Updating Detailed Engineering Designs and Financing Option Study

Year: February 2012 to December 2013

Location: India

Client: Wilbur Smith Associates Private Ltd

Main project features: The government of Kerala is in the process of upgrading about 367 km of state roads under the Kerala State Road Projects Phase-II (KSTP-II), through a loan from the World Bank. The government required consultancy services for updating detailed engineering designs and the financing options available to the state to fund these projects. CRIS is executing this assignment, as part of the consortium with Wilbur Smith Associates Private Ltd. . CRIS has the responsibility to study various financing options available and their *pros* and *cons* and analyse feasibility for each of these modes for each stretch under KSTP-II.

Position Held:Project Leader

Activities performed: Prepared the framework of the study to be carried out by a team of four people, participated in all the client interactions and presentations, and reviewed the output quality

80. Name of the assignment: Assistance in conducting PPP risk allocation workshop by roads sector expert

Year: March 2012 to June 2012

Location: Indonesia

Client:Indonesia Infrastructure Guarantee Fund (IIGF)

Main Project Features:

The scope of work and approach for this assignment have been divided into two broad tasks:

- Pre-work for risk allocation workshop
- Assisted in conducting workshop

Pre-work for the risk allocation workshop:

- Prepared a context-setting presentation for each sectoral discussion
- Provided necessary input for the project case (1-2 page brief write-up supplemented with a

draft PPP agreement) prepared by the IIGF team

- Provided a template for Key Provision for PPP agreement in the selected sector, based on international best practice, as to enhance understanding of risks
- Allocated and drafted the PPP agreement

Assistance in conducting workshop:

- Aided the IIGF team in managing flow of the discussion to get maximum value
- Share CRISIL's knowledge and experience in PPP and risk-sharing in relevant infrastructure sector
- Prepare/deliver concluding remarks for each panel discussion

Position held: Project Leader

Activities performed: Supervision, review of quality of output

81. Name of the assignment: Assisted in preparing bid for the Jalgaon-Gujarat Border Stretch of the NH-06 developed under NHDP Phase III, NHAI

Year: March 2012 to May 2012

Location: India

Client: BSCPL Infrastructure Ltd.

Main project features: The client wished to participate in the bid called by NHAI and sought CRIS's help for the same. The scope of the project included:

- Traffic study involving seven-day classified volume count and 1-day OD count at three locations that led to the assessment of the average annual daily traffic and projection of the same over the concession period
- Performed industry analysis and scenario mapping to evaluate impact of industry development in the project influence area and developed scenarios
- Financial modelling incorporating the concession agreement, traffic study, and scenario analysis and recommending a bidding strategy to the client

Position held: Project Leader

Activities Performed: Prepared the framework of study to be carried out by a team of four people and participated in all client-interaction presentation

82. Name of the Assignment: Financial analysis for 6-laning of Paradeep-Chandikhole stretch of NH-5A Project in Orissa on a build-operate-transfer (BOT Toll) DBFOT pattern under NHDP Phase V, NHAI

Year: March 2012 to April 2012

Location: India

Client: Unity Infrastructure Assets Ltd.

Main project features: The client wished to participate in the bid called by NHAI and sought CRIS's help for the same. The scope of the project included a financial analysis incorporating the concession agreement, traffic study, and scenario analysis and recommending a bidding strategy to the client.

Position Held: Project Leader

Activities performed:

- Overview of the project corridor
- Base year traffic assessment through field surveys
 - Classified traffic volume count
 - Origin-destination surveys
 - Intersection turning movement counts
- Traffic data analysis
 - Classified traffic volume count
 - Categorized break-up of motorized traffic
 - Calculation of annual average daily traffic (after seasonality corrections)
 - Origin-destination surveys
 - Travel characteristics and pattern
 - Trip generation and termination points on corridor
 - Intersection turning movement counts
 - Intersection levels with the project corridor
 - On- and off-ramp movements from the project corridor
- Traffic projections:
 - Socio-economic data analysis
 - Economic and traffic growth rates
 - Estimation of projected toll-able traffic for concession period

83. Name of the Assignment: Traffic study and financial modeling for four-laning of Kundapur to Karnataka and Goa border stretch of NH-17

Year: March 2012 to April 2012

Location: India

Client: Unity Infrastructure Assets Ltd.

Main project features: The client wished to participate in the bid called by NHAI and sought CRIS's help for the same. The scope of the project included a financial analysis incorporating the concession agreement, traffic study, and scenario analysis and recommending a bidding strategy to the client.

Position Held: Project Leader

Activities performed:

- Overview of the project corridor
- Base year traffic assessment through field surveys
 - Classified traffic volume count
 - Origin-destination surveys
 - Intersection turning movement counts
- Traffic data analysis
 - Classified traffic volume count
 - Categorized break-up of motorized traffic
 - Calculation of annual average daily traffic (after seasonality corrections)
 - Origin-destination surveys

- Travel characteristics and pattern
- Trip generation and termination points on corridor
- Intersection turning movement counts
 - Intersection levels with the project corridor
 - On- and off-ramp movements from the project corridor
- Traffic projections:
 - Socio-economic data analysis
 - Economic and traffic growth rates
 - Estimation of projected toll able traffic for concession period

84. Name of the assignment: Bid advisory on Dewas-Shivpuri Road

Year: 2011-2012

Location: India

Client: VINCI Concessions Development III Pte. Ltd.

Main project features:

The detailed scope of work was:

Review and risk assessment of the concession agreement (CA) and related documents

Review of documents

- Reviewed the CA and documents including the toll policy and identify issues having financial/commercial/legal/contractual implications for the bidder
- Reviewed publicly available information about recent bidding processes to understand the trends in such transactions, specifically ones which have financial implications for the bidder
- Identified critical risks, specifically in the areas of termination, competition, and exit
- Listed critical assumptions to be factored in the financial model based on the critical assessment of the CA

Financial modelling

- Preparing a comprehensive financial model would include:
 - Listing assumptions that are required to be inputs in the model for the project.
 - Preparing the financial model, free of date macros, based on the assumptions and the terms and conditions identified in the preceding phase.
 - Projecting project financials including revenue, expenditure, cash flows, assets and depreciation
 - Working out financial statements (including those on profit & loss, cash flow and balance sheet), and computation of key ratios such as the IRR and DSCR
 - Identify key input parameters affecting the output and their spread
 - Carryout a stand lone sensitivity analysis as well as a multi-variate Monte Carlo simulation based on identified input parameters
 - Suggesting a capital structure that shall improve the bankability of the projects from the point of view of the potential financiers to the project
 - Separate IRR calculations for HCON and VINCI

Position held: Project Leader

Activities Performed:: Supervision, review of quality of output, key interface for the client

85. Name of the Assignment: Preparing financial model for road (BOT and annuity) Projects

Year: June 2011

Location: India

Client: GKC Projects Ltd

Main project features: The client wished to develop a comprehensive financial model to assist in its bidding process for NHAI Road Projects, both annuity and toll. The scope of the assignment included:

- Reviewing concession agreements and key assumption parameters to be considered for financial modelling
- Preparing financial model incorporating features
- Develop two/four/six lanes with options for up to six toll plazas
- Different growth rate for toll plazas
- Input parameters to be in a single sheet
- Provision of sub-debt/refinancing options
- Provision for dividend distribution
- Flexibility to incorporate provisions for local traffic
- Option for scheduling of major maintenance
- Other options as per the concession agreement

Position held: Project Leader

Activities Performed: Prepared framework of the study to be carried out by a team of four people and participated in all client interactions and presentations

86. Name of the Assignment: Bid advisory for Shivpuri-Dewas Road

Year: June 2011 to August 2011

Location: India

Client: HCC Concessions Company Ltd

Main project features: During the study, the team undertook the following tasks:

- Review and risk assessment of the concession agreement (CA) and related documents
 - Review of Documents
 - Review the CA and documents including toll policy and bring out issues having financial/commercial/legal/contractual implications for the bidder.
 - Review publicly available information about recent bidding processes to understand the trends in such transactions, specifically ones which have financial implications for the bidder.
 - Identify critical risks, specifically in areas of termination, competition, and exit.
 - List of critical assumptions to be factored in the financial model based on the critical assessment of the CA
- Financial modelling

Preparing a comprehensive financial model would involve the following tasks:

- Listing assumptions that are required to be inputted in the model for the project
- Preparing financial model and free of date macros, based on the assumptions and the terms and conditions identified in the preceding phase

- Forecasting project financials including revenue, expenditure, cash flows, assets and depreciation
- Preparing financial statements (including profit & loss, cash flow, and balance sheet), and computation of key ratios such as the IRR and DSCR
- Identifying key input parameters affecting the output and their spread
- Carrying out a standalone sensitivity analysis as well as a multivariate Monte Carlo simulation based on identified input parameters
- Suggesting a capital structure that shall improve the bankability of the projects from the point of view of the potential financiers to the project Separate IRR calculations for HCON and VINCI
- Recommending the bid strategy

This task shall involve the following activities:

- Consideration of the risk and financial results as generated in the previous tasks
- Assessment of the competition for the bidding process
- Recommendation of the bid strategy and final presentation

Position held: Project Leader

Activities Performed: Supervision, review of quality of output, key interface for the client

87. Name of the Assignment: Preparing a revised RFP for selecting a financial consultant

Year: March 2010 to May 2010

Location: India

Client: National Highways Authority of India (NHAI)

Main project features: NHAI had a long-term financial investment plan for developing national highways. They had also identified funding sources for this purpose. Based on their plan, it was estimated that around Rs 15,000 crore would need to be raised by NHAI through long-term debt from the market. NHAI proposed the appointment of an advisor to assist it in preparing a fund mobilization plan to raise the planned resources.

With this background, NHAI appointed CRIS to assist it in preparing the RFP, including defining the qualification and evaluation parameters for the appointment of the financial advisor

Position held: Team Member

Activities Performed: The scope of services included the following:

- Vetting terms of reference for the financial advisor
- Determination qualification and evaluation parameters
- Finalisation of RFP for appointing financial advisor
- Assistance in responding to queries of the bidders

88. Name of the Assignment: Pre-feasibility and financial advisory services for construction of an Inter Modal Transit Centre at Byappanahalli, Bangalore

Year: 2009

Location: India

Client: Karnataka State Road Transport Corporation (KSRTC)

Main project features: The KSRTC proposes to construct a state-of-the-art bus terminal (inter-modal transit centre) with passenger amenities at Byappanahalli over 9 acres obtained from the NGEF. This bus terminal will facilitate the operations of buses plying to the North of Bengaluru city. The proposed Inter-Modal Transit Centre will be provided with bus terminals, passenger amenities, and other transport related facilities on a PPP basis with viability-gap funding.

Position held: Project Leader

Activities performed: The scope of consultants' scope of work includes conducting pre-feasibility studies and providing financial advisory services for projects.

89. Name of the assignment: Transaction advisory services for proposed Kigamboni Bridge project to be implemented under PPP mode

Year: October 2009

Location: Tanzania

Client: National Social Security Fund, The United Republic of Tanzania

Main project features: The client wished to construct the Kigamboni bridge in Dar es Salaam, Tanzania. CRIS was mandated to assist in providing transaction advisory services for the Kigamboni bridge project

Position held: Team Leader

Activities performed: The areas of assistance included the following:

Phase 1:

- Reviewed project preparation including feasibility studies carried out and also liaised with the National Social Security Fund (NSSF), the Ministry of Infrastructure Development, National Environmental Management Council, Public Procurement Regulatory Authority and Tanzania Ports Authority
- Updated studies and proposed project business model to clarify the role of the PPP partner
- Evaluation of best PPP structure, based on a qualitative or quantitative assessment of value-for-money
- Reviewed and/or development of a comprehensive risk matrix for the PPP project
- Reviewed the cost estimates and draft designs
- Reported on all activities carried out under Phase 1 that summarize the main conclusions of the TA and described documents developed in Phase 1.

Phase 2:

- Developed Gantt chart for project procurement activities to assist the procurement process
- Prepared a definitive and realistic timeline for procurement
- Prepared of a detailed RFP and the draft PPP agreement, preparation of the final RFP following consultations with the prospective bidders
- Participated in the procurement committee for the evaluation of bids along with other technical advisors as needed
- Assisting the NSSF in negotiation with the preferred bidder
- Participating as an advisor in the selection committee to evaluate bids
- Assistance in the negotiations and drafting of the final PPP contract

F. AVIATION:

90. Name of the Assignment: Project Consultant for Kannur International Airport Project

Year: November 2012 to October 2014

Location: India

Client: Kannur International Airport Limited

Main project features: The government of Kerala (GoK) proposes to establish a greenfield international airport at Kannur. The GoK has selected the site and has obtained the Centre's in-principal approval to establish the proposed airport. Kannur International Airport Limited (KIAL) has been incorporated as a public limited company that will develop and manage the project. KIAL selected a consortium of AECOM and CRISIL through an open competitive bidding process as project consultants for the airport's development. The consultants will be handholding KIAL to develop the project from the design to the operational stage and provide strategy, technical, finance (excluding financial closure), legal, and operational consulting support.

Position Held: Project supervisor

Activities Performed: The scope of services being provided included:

- Review of DPR with a particular reference to the project's financial analysis, financial viability and sustainability, to advise KIAL on the adequacy of the DPR
- Revision of financial analysis such as project benefits (revenue/income sources) project costs (viz., capital investment, routine maintenance cost, periodical maintenance cost, operating costs, etc.), projected cash flow statements, profit and loss statements, depreciation schedule for various capital assets, financial institution loan repayment schedule, break-even analysis, risk analysis, etc, based on airport usage as suggested by the traffic analysis and other data for the analysis period, broken down to construction period and operation period and other potential data as contained in the DPR, if required, based on the consultant's assessment of DPR
- Preparing an alternate financial plan with capital outlay, debt-equity ratio, break-even analysis, risk analysis and other financial indices and ratios to showcase the project's viability to the potential investors/bankers, if required, based on the consultant's assessment of DPR
- Financial restructuring of the project with viable means of raising the project funds and structuring of equity and debt/loan portfolios, if required, based on the consultant's assessment of DPR
- Assisted in locating investors, planning of fundraising methodology and implementation, corporate structuring, and overall assistance in implementation
- Prepared an optimum business plan advising the airport operator on the best possible business model to run the airport for each of the cost/profit centers

91. Name of the assignment: Advising the Union Ministry of Civil Aviation, Government of India, on development fee (DF) and user development fee (UDF) at selected non-metro airports

Year: June 2011 to December 2011

Location: India

Client: Airports Authority Of India

Main project features: The study involved:

- Determination of DF/UDF based on the aero asset base of the airport and its funding, depreciation and cost of capital, projected revenues, projected operating expenditure, etc.
- Due diligence, traffic projections, etc.

Position held: Project leader

Activities performed: Supervision, review of quality of output, key interface for the client.

92. Name of the assignment: Estimating cost of equity

Year: January 2011 to May 2012

Location: India

Client: Mumbai International Airport Private Limited (MIAL)

Main project features: The MIAL is interested in estimating its cost of equity for which CRIS had provided assistance.

Module 1: Review of existing literature and benchmarks

This module included:

- Review of existing literature available with the client dealing with the evaluation of cost of equity for MIAL
- Review of existing best practices used by similar global entities to compute cost of equity
- Benchmarking of 3-4 infrastructure projects in India and airport projects elsewhere

Module 2: Estimation of cost of equity

Under this module, the cost of equity for MIAL was assessed through two approaches:

- Capital asset pricing model (CAPM)
- Arbitrage pricing model (APM)

Position held: Project Leader

Activities Performed: Supervision, review of quality of output, key interface for the client

93. Name of the assignment: Consultancy services in preparation of compendium of central govt services and regulations for greenfield airports

Year: October 2010 to February 2012

Location: India

Client: Airports Authority of India (AAI)

Main project features: The study analyzed:

- Details of services to be provided by the government agency to the private operator for a greenfield airport
- Personnel, equipment, and infrastructure requirements of that the government agency requires for providing services
- Back- and front-end space requirement of the government agency for providing their services
- Cost recovery mechanism to be adopted by the government agency
- Agreement/MoU to be signed by the government agency with the airport operator for services to be provided
- Guidelines for airport infrastructure requirement in terms of specifications, for (a) airside infrastructure (like runway, apron, and taxiways) and (b) terminal building for different categories of airports
- Directions for safety and security requirements

Position held: Project Leader

Activities Performed: Supervision, review of output quality, key interface for the client

94. **Name of the assignment: Advisory Services to Assist PT PP in Understanding and Analyzing India's Experience in Airport Privatization**

Year: November 2011 – March 2012

Location: Jakarta

Client: PT. Pembangunan Perumahan (Persero) Tbk

Main project features:

Study tour to India/identification of a strategic partner:

- Gaining practical knowledge/first-hand experience of India's privatized airports
- Engaging in discussions with the various stakeholders
- Assessing compatibility of some private airport operators

Negotiation for structuring joint venture (JV) and closure of the transaction:

- Based on understanding of the JV criteria, suggesting possible investment options with the JV partner and also highlighting key issues governing the transaction these include:
- Key commercial parameters for the JV
- Capital structuring options
- Shareholding pattern, related rights and obligations
- Mechanics and merits/demerits of various structures
- Other commercial and financial parameters under various structuring options

Position held: Project Leader

Activities performed: Supervision, review of output quality, key interface for the client

G. LOGISTICS:

95. **Name of the Assignment:** Appointment of Consultant for market study, Macro & Micro analysis, preparation of feasibility report (based on detailed financial model), Business Plan & Project Concept and Implementation Road Map for development of a Logistics Park.

Year: July 2021- June 2022

Location: Mumbai India

Client: RattanIndia Power Limited

Main project features: The Assignment was conducted to prepare business plan for logistics facilities in Nashik & Amravati districts of the state of Maharashtra . The key activities were:

1. Module-1: Macro Market Overview
 - 1.1. India Market Overview
 - 1.2. Logistics sector in India including transportation, warehousing, industrial and food parks in India
 - 1.3. Recent transactions in logistics parks (investors, target overview, typical valuation, Gross IRRs, investment thesis etc.)
 - 1.4. Regional Market Overview, covering:

	1.5. Site overview 2. Module-2: Micro Market Overview Maharashtra Market Overview 2.1. Primary catchment/ influence area (IA) overview (for 1 land parcel) 2.2. Competitive landscape for select locations 3. Module-3: Project Concept 3.1. Similar case studies for benchmarking 3.2. Strategic Options for the land parcel 3.3. Option shortlisting 4. Module-4: Financial Model and Implementation Plan 4.1. Financial Plan for the project Including preparation of : 4.1.1. Balance sheet 4.1.2. Profit & Loss statement 4.1.3. Free Cash flow 4.1.4. Ratios and IRR 4.2. Risk mitigation and implementation plan 4.2.1. High level internal and external risk assessment and mitigation plans Position Held: Team Leader Activities Performed: Supervision, review of quality of output
96.	Name of the Assignment: Business Plan for Goods Logistics Business for MSRTC Year: April 2018- June 2019 Location: Mumbai India Client: Maharashtra State Road Transportation Corporation (MSRTC) a state government entity , KPMG Main project features: Business Plan Development for Goods Logistics Business Logistics that included : 5. Market Overview 6. Maharashtra Market Overview 7. Analysis of Existing Infrastructure 8. Legal Provisions 9. Options for Business Model 10. Financial Analysis 11. Business Structure 12. Opportunities with Mega Infrastructure projects 13. Recommendations Position Held: Team Leader, Sector Expert Activities Performed: Supervision, review of quality of output
97.	Name of the Assignment: Market due diligence for CFS-ICDs and private freight terminals (PFT)

Year: October 2015- Feb 2016

Location: India

Client: International Finance Corporation (IFC) and Continental Warehousing Corporation (CWC)

Main project features: The study involved market assessment of three CFSs and six PFT-cum-ICD facilities to understand the market potential for these. CFSs are located at Mumbai, Chennai & Tuticorin, while PFT cum ICD facilities are located at Hyderabad, Ahmedabad, Panipat, Chennai, Bengaluru and Kolkata. The study was intended for an investor who wants to assess the market potential of the facilities across various locations.

Position Held: Team Leader

Activities Performed: Supervision, review of quality of output

98. Name of the assignment: Techno-economic viability study for Khurja Facility

Year: February 2014 to February 2016

Location: India

Client : Arshiya Northern FTWZ Limited

Main Project Features: The client aims to restructure its existing facilities operated by subsidiary companies at Khurja. Currently, two of the client's wholly-owned subsidiaries operate an integrated facility in Khurja comprising a 51-hectare FTWZ by Arshiya Northern FTWZ Ltd., and a 26 hectare logistics park by Arshiya Industrial and Distribution Hub Limited.

- Operations of these facilities were severely affected owing to regulatory issues. Hence, the parent company, Arshiya Ltd aims to restructure the existing facilities by merging existing facilities of AIDHL and ANFL into Sector specific SEZs and an ICD into Arshiya Transport & handling Ltd (ATHL). The scope of work for the assignments includes:
- Market assessment
- Regulatory assessment
- Financial and economic assessment

Position Held: Project Leader

Activities Performed: Supervision, review of output quality, key interface for the client

99. Name of the Assignment: Study of freight rates in the trucking industry

Year: December 2012 to March 2013

Location: India

Client: Hindustan Unilever Ltd (HUL)

Main Project Features: The client is one of India's largest consumer goods company based in Mumbai, Maharashtra. Its products include foods, beverages, cleaning agents and personal care products. It is a subsidiary of the British-Dutch company Unilever, one of the world's leading suppliers of fast-moving consumer goods with a presence in more than 100 countries. HUL has over 35 brands in its portfolio spanning 20 distinct categories such as soaps, detergents, shampoos, skin care, toothpastes, deodorants, cosmetics, tea, coffee, packaged foods, ice cream, and water purifiers. HUL wishes to undertake a market survey to understand the trends in freight rates in the Indian trucking industry, impact of fuel prices and outlook for the same.

Position held: Project Leader

Activities performed: The scope of work included primary and secondary research:

- Brief overview of Indian trucking industry
 - Value chain of services
 - Type of players
 - Key industry characteristics
- Analysis of trends in truck freight rates for identified origin-destination pairs
 - Trends in truck freight rates before and after the hike in diesel prices (around September 2012 when the change was affected)
- Outlook for trucking industry based on analysis of freight rates
 - Short-term outlook on movement of freight rates (H1 of 2013)
 - Medium-term outlook on movement of freight rates (H2 of 2013)
 - Broad-level long-term outlook (up to 2014-15)
 - Analysis of impact of expected changes in the operating environment of trucking industry such as the Goods & Services Tax (GST).

100. Name of the assignment: Consultancy services to FCI for constructing silos

Year: November 2012 – November 2013

Location: India

Client: Food Corporation of India (FCI)

Main project features: The Food Corporation of India (FCI) is one of the premier government organizations set up under the Food Corporations Act, 1964, to fulfill the following objectives of the Food Policy:

- Effective price support operations to safeguard farmers' interests
- Storage, preservation and supply of foodgrains throughout the country for public distribution system and other government schemes
- Maintaining a satisfactory level of operations and buffer stocks of foodgrains to ensure national food security

Position held: Project Leader

Activities performed: The scope of services being provided by CRIS includes:

- Preparing a Manual of Standards and broad specifications for silos for storing foodgrains
- Preparing bid documents, Drafting and assisting in finalising the NIT/bid document, including appropriate legal and operational covenants, which incorporate both domestic and international best practices and conform to the requirements of the Central Vigilance Commission (CVC) and other applicable guidelines
- Assisting in floating an RFQ in the market and shortlisting qualified parties;
- Assisting in the bid process management, launching the RFP, formulation of responses to bidder queries, bid evaluation and recommendations per criteria in the RFP document, recommending a private developer, contract negotiations and bid closure, which are required to be undertaken for achieving technical close
- Launching the bid process including coordinating appropriate pre-bid conferences, development of a bidder shortlist, providing assistance in the evaluation of bids leading up to final bid award
- Drafting a model concession agreement as part of the bid and contract framework to be signed between FCI and the successful bidder and drafting a commercial document that would

adequately ensure safeguarding the FCI's interests of FCI and its assets over the entire concession period

- Finalising project structuring issues -including payment mechanisms, development of performance parameters for the PPP operator, institutional and legal structuring issues, finalization of a risk matrix, legal documentation, compliance and contracting issues

101. Name of the assignment: Traffic study and financial assessment of cold chain business

Year: November 2011 to March 2013

Location: India

Client: Safe & Secure Logistics Pvt. Ltd (SSLPL).

Main project features: The client plans to expand its existing transportation business , including temperature-controlled transportation, into the cold storage business to offer integrated cold chain services. The scope of projects included:

- Study of extant cold storage operations of the client in terms of major commodities, routes and seasonality
- Analysis of growth potential of traffic in the cold storage segment for the next five years, including existing and new commodities across existing and new routes
- Competition analysis of major and new players (five competitors)
- Estimation of traffic for the client, encompassing development of scenarios under different assumptions and projections for the future 5 years
- Estimation of capacity expansion required for the client
- Providing broad estimates for capacity expansion to be undertaken by SSLPL for its cold chain operations

Position held: Project Leader

Activities performed: Prepared the framework of study to be carried out by the team, participated in all the client interactions and presentations, supervised and reviewed output quality, and acted as the key interface for the client.

102. Name of the Assignment: Presentation on the Indian Logistics Industry

Year: February 2012- February 2012

Location: India

Client: Eredene Infrastructure Private Limited

Main Project Features: The client, a major private equity player investing in the logistics space, wished to have an overview of the logistics industry in India, regulatory environment, investment scenario, and development trends and also economics of operations. The objective was to identify opportunities for investment and provide an overview to the internal team.

The detailed presentation covered 10 segments of the logistics industry and focused on 4 segments, viz. parcel cargo, rail logistics, express cargo and cold chain. In the focus segments, the economic aspects for a player were dealt in greater detail and the assessment parameters were highlighted.

Position held: Project Leader

Activities performed:

- Taxation or technical services
- Audit and accounting services
- Roadshows

- Liaison with statutory/local authorities, government departments, etc.
- Primary surveys
- Due diligence of any specific asset

103. Name of the assignment: Validation of market potential and financial feasibility of an ICD and PFT in Ahmedabad and Delhi NCR Market

Year: December 2011 to February 2012

Location: Mumbai, India

Client: Continental Warehousing Corporation (NhavaSheva) Limited

Main Project Features: Continental Warehousing Corporation (NhavaSheva) Limited (CWCNSL) is planning to set up multi-modal logistics parks consisting of an ICD and a PFT at Panipat and Ahmedabad. CWCNSL mandated CRIS to undertake a study to evaluate the viability of this business.

Position Held: Team member

Activities performed: The activities performed on this assignment include the following:

Industry analysis of:

- Indian container and bulk cargo market in Gujarat with particular focus on Ahmedabad region and in the Northern Capital Region with focus on Panipat region
- Major players - margins - capacities - customers - growth clocked etc. -
- Risks and critical success factors

Project Viability Analysis

- Project catchment area analysis – Primary hinterland and secondary hinterland traffic potential
- Determination of potential volume (including seasonality) and expected growth for container/bulk cargo
- Analysis of competing facilities/alternatives available - Comparison of cost/time to end user
- Total logistics cost analysis for using companies' facilities vis-à-vis other facilities justifying traffic migration to proposed facilities
- Determining comparative advantage of using proposed facilities (business model/location advantages/disadvantages/connectivity to ports and internal hinterland)
- Tariff comparisons with existing competing ICDs in the catchment area based on interactions with 5-6 users of the facilities
- Analysis of future threats - Emerging competition and disruptive events in the immediate/long term

Project validation

- Capital cost benchmarking based on publicly available information and limited primary interactions (2-3) with relevant personnel
- Cost validation, capacity confirmation and project implementation schedule validation
- Operating cost, margin for proposed facilities and margins for existing facilities in the catchment area shall be based on publically available information & limited primary interactions

104. Name of the assignment: Study of the Express Services Industry

Year: December 2011 to October 2012

Location: India

Client: Express Industry Council of India

Main project features:

- Current size and growth drivers
- Industry structure
 - Segmentation based on the value chain of services
 - Classification of players based on size
 - Profiling of major players and end-users
- Study of operating environment
 - Assessment of competitive environment
 - Identification of major issues concerning the industry players
 - Analysis of recent developments in the industry
- Analysis of service quality (user perspective), included identification of:
 - Service parameters
 - Appreciated service parameters in an express service
 - Disapproved or disliked service parameters in an express service
- Analysis of industry outlook
 - Analysis of growth drivers
 - Determination of the projected size of the industry
 - Scenario analysis
 - Major growth segments in the industry and success factors
 - Broad assessment of employment growth due to growth in the express services industry
 - High-level assessment of the likely investment in the sector

Position held: Team Leader

Activities performed: Led a team of two members in preparing the industry study, was involved in key interactions with various express logistics players, and was responsible for the overall quality of the output

105. Name of the assignment: Comprehensive commercial due diligence of a port opportunity for a private equity player

Year: 2011

Location: India

Client: A private equity player

Main project features: CRIS assisted the client in conducting a comprehensive due diligence for investment in the assets. The assignment involved assistance in the contractual review of the concession agreements, market and traffic assessment, development assessment, risk assessment, financial modelling, and holistic evaluation of the potential investment opportunity from the commercial and market perspective.

Position held: Team Leader

Activities performed:

- Auditing of prepared financial models
- Contractual diligence
- Undertook market and traffic assessment
- Development and risk assessment
- Financial modelling
- Holistic evaluation of the potential investment opportunity from the commercial and market perspective
- Delivery of all these modules prepared in the form of review note to the client

106. Name of the assignment: Advisory assistance for bidding for development of logistics hub at Pawarkheda, Madhya Pradesh

Year: May 2011 to July 2011

Location: India

Client: Punj Lloyd Infrastructure Ltd.

Main project features: The study involved assessment of:

- Market for proposed infrastructure facilities and services
- Financial feasibility of the project
- Commercial and contractual risks inherent in the project and the draft
- Authorization agreement

Position held: Project Leader

Activities Performed: Supervision and review of the output quality, key interface for the client

107. Name of the assignment: Viability study for chemical logistics in India utilizing tank containers

Year: December 2010 to August 2011

Location: India

Client: Fourcee Infrastructure Equipment Pvt Ltd

Main project features: Fourcee Infrastructure Equipment Pvt. Ltd., a fast-growing logistics company, is focused on the movement of liquids, with special emphasis on non-petroleum oil and lubricants (non-POL) such as fatty acid, crude palm oil, molasses and chemicals, by rail in containers hitherto transported by road in tank containers. Fourcee's value proposition relies on its competitive costing and seamless movement of cargo in ISO containers. In short, it proposes to provide a cost-effective and efficient door-to-door movement of specialized cargo to its customers. Forsee currently handles around 1 million tonnes of liquid cargo. To increase its volume, Fourcee proposes to purchase/lease more containers under the current Container Train Operations (CTO) policy. Also, it has applied for the Special Freight Train Operator (SFTO) licence. Fourcee has appointed CRISIL Infrastructure Advisory to conduct a viability study of its future businesses

108. Name of the assignment: Study on business plan for multi-modal logistics park (MMLP) at Khodiyar

Year: 2010

Location: India

Client: Container Corporation (CONCOR)

Main project features: The salient aspects of study were:

- Identification of industrial clusters in the catchment area through secondary data sources of, interactions with government agencies and other stakeholders, etc.
- Cargo identification, volume and growth assessment of potential cargo that could be handled at the proposed MMLP
- Determination of capital expenditure to handle the identified cargo
- Assessment of operational expenditure such as payment to railways, operation and maintenance costs, payment to road transportation, electricity costs, manpower cost, depreciation, etc.
- Revenue analysis including freight, warehousing, value-added logistics services, handling revenue, etc.
- Preparation of a bankable business plan
- Sensitivity analysis

Positions held: Project Director and Team Leader

Activities Performed: Supervised study on the business plan

109.

Name of the assignment: Business

feasibility for air cargo logistics in Hyderabad for Inlogistics innovate B2B

Year: 2010

Location: Hyderabad, India

Client: Inlogistics B2B Private Limited

Main project features: The study encompassed:

- Demand analysis
 - As-is commodity analysis for export-import (EXIM) and domestic cargo
 - Catchment area assessment for Mamidipally CFS
- Supply analysis
 - Competition analysis: Identifying and profiling players
 - Operational aspects of various existing CFS
 - SWOT analysis
 - Proposed CFS venture in the area
- Market mapping
 - Value-chain understanding of business
 - Mapping of export and import-business cycle
- Business case development
 - Key success factors for Mamidipally
 - Infrastructure assessment
 - Cost assessment
- Financial Planning and Bidding Strategy
 - IRR and sensitivity analysis
 - Determination of revenue share/grant etc.
 - Business intelligence

Position held: Project Director

Activities Performed: Managed the entire study, suggested the course of action, and advised on the bidding strategy

110.

Name of the assignment: Logistics study

for Mumbai for one of private container operators' innovative B2B

Year: 2009

Location: Mumbai, India

Client: A private container operator

Main project features: The study included:

- Collecting data necessary to ascertain market in terms of overall size, commodity-wise (tonnage-wise) break-up, profile of vehicles, tonnage carried by vehicles forming critical O-D pairs with sufficient volumes, getting customer information, transport agencies, rates, etc.
- Arriving at market wise detailed break-up in terms of requisite variables (tonnage, commodity, pricing), triangulation with macro data, internal brainstorming on findings, fine-tuning, quantitative assessment, conclusions, etc.
- Independently verifying the findings of O-D data and analysis, obtaining market insights in terms of pricing, handling, movement of goods (in-transit places, dwell time), and better appreciation of road transport offerings. Other objectives include identifying gaps in current offerings and latent needs to the market; understanding the value chain in terms of relative importance of freight forwarder, warehouse agents, transporters, shippers and other players; gauging if there is a specific point of negligence; And collating specific intelligence on commodity/region/issues/pain points for each company to map the complete industry structure
- Undertaking an extensive and critical supply-side analysis of competitors in domestic rail space including Indian Railways & Concor to assess their investments, assets, volumes, share in the space, strategies, location, links in the value-chain and value-proposition offered, thereby evaluating where exactly the client can provide value-add
- Collecting commodity-wise rail data for a week to obtain the modal share of road and rail for EXIM traffic on the long haul
- Preparing a comprehensive B-Plan for the client keeping in mind the strategic assets, medium to long term plan, and SWOT analysis of the other players. This included forecast of both the domestic and EXIM market (commodity-wise) in 3-year intervals along with break-up of road-bulk, road-break bulk, road-containerized, and similar forecast for railways
- Determining market share for the client after discussions with its representatives on pricing, asset creation, services offerings, and other elements of the B-Plan

Position held: Project Director

Activities Performed: Defined the project's key contours in discussion with the client, managed the study from commencement to conclusion, reviewed all the deliverables, and participated in all the client interactions and presentations.

111.

Name of the assignment: Advise Reliance

Anil DhirubhaiAmbani Group, a leading business house for building their container train logistics business

Year: 2009

Location: Mumbai, India

Client: A leading business house

Main project features included:

- Preparing the roadmap and assisting the client-side project team in developing processes and systems to commence operations
- Suggesting organization structure and the manpower needed to support business
- Suggesting revenue model for the business
- Providing inputs on MoU entered with partners
- Undertaking due diligence on the MoU partners
- Devising investment strategy
- Developing a matrix to support the management's decision-making efforts

Position held: Team Leader

Activities Performed: Led efforts to advise this business house for building their container train logistics business and facilitated to run the first load of rake.

112. Name of the Project: Land-side development study for a global container port operator to identify container logistics related opportunities across India with focus on Mumbai and Chennai.

Year: December 2011 to July 2012

Location: Mumbai

Client: Port of Singapore Authority (PSA)

Main Project Features: The client who is a subsidiary of one of the world's leading port operators, which also runs container terminals in many major ports, was keen to explore land-side development opportunities to support and complement its port operations. In this regard, it wanted a land-side developments study to be undertaken and appointed CRIS for the same.

Position held: Project Leader

Activities performed: The scope of work was divided into:

- Analysis of land-side developments done by the client's key competitors in terms of scale, scope, spread and broad profitability
- Market assessment of various land-side developments based on cargo movement. Identification of suitable projects such as ICD/CFS/Logistics park/tank farms, etc.
- Assessment of existing facilities for the land-side developments identified

H. CONTAINERS

113. Name of the assignment: Land-side development study for a global container port operator to identify container logistics related opportunities across India with a focus on Mumbai and Chennai

Year: December 2011 to July 2012

Location: India

Client: PSA India Pte. Ltd.

Main Project Features: The client, one of the largest container port operators globally, is keen to explore land-side opportunities in India and mandated CRISIL to undertake a market assessment study on the same.

Position held: Project Leader

Activities Performed: The study involved:

- Analysis of land-side developments of PSA's key competitors in terms of scale, scope, spread, and broad profitability
- Market assessment of various land-side developments based on cargo movement and identification of suitable projects such as ICDs, CFSs, logistics parks and tank farms
- Assessment of existing facilities for identified land-side developments. The study will be conducted using a combination of primary surveys, primary interactions with 25-30 stakeholders such as end users and intermediaries (CHA, freight forwarders, 3PL players), and secondary desk research.

114. Name of the assignment: Preparation of entry strategy for an established logistics player to set up ICD/CFS in north-west hinterland of India

Year: September 2007 to December 2007

Location: India

Client: Balmer Lawrie

Main project features: The study included:

- Assessment of container volumes at the competing ports in the north-west region that included JNPT, MbPT, Kandla, GPPL, and GAPL (now Mundra)
- Forecasting of the container volume for each of these ports along with the analysis of capacity utilisation, capacity additions, infrastructure improvements, logistics cost advantage from cargo generating centres, etc.
- Mapping existing competition for CFSs to understand capacity, pricing, and services offered by them and also assessing the key success factors such as linkages with shipping lines
- Assessing other dynamics at the port side, such as capacity available with the import and proportion of import and export traffic
- Based on above, estimating the targeted volume for the client
- Determining investment required to handle the above estimated volume
- Determining break-even volume and revenue based on the *pro forma* financials

Positions held: Project Leader & Team Member

Activities Performed: Responsible for managing the entire study and quality of deliverables and also was the key interface with the client in the project

115. Name of the assignment: Business strategy for a new entrant in container cargo business

Year: April 2007 to August 2007

Location: India

Client: KrishakBharati Cooperative Limited (KRIBHCO)

Main project features: The study's salient features were:

- Analysis of container cargo logistics for north-western hinterland that included volume and modal breakup for each origin-destination pair
- Competitor analysis in terms of investment planned, routes plied, etc
- Assessing the impact of discontinuities such as a dedicated freight corridor, impact of the Golden Quadrilateral, etc.
- Development of four scenarios for the client, based on external changes in external environment and internal capabilities
- Suggesting the most suitable strategy keeping the risks and return in consideration. The strategy included the market choice, routes to be plied upon, investment phasing, etc.
- Other aspects included assessment of leased and owned rakes. The outcome helped in fine tuning the strategy.
- Also developed a detailed economic logic to assess conversion of existing fertilizer rakes vs. utilizing standard flat wagons
- Detailed financial modelling to quantify impact of the abovementioned aspects

Position held: Project Leader

Activities Performed: Led the study with a team of two members and was responsible for all the modules of the study and the quality of output

116.

Name of the assignment: Corporate

strategy for Container Corporation of India Ltd. (CONCOR)

Year: 2005 to 2006

Location: India

Client: Container Corporation of India Ltd. (CONCOR)

Main project features: The study included:

- A study of CONCOR's current business model and internal processes and suggesting a strategy for business process reengineering to suit the changing business scenario
- Suggestions on future expansions of CONCOR's operations in rail, road, and port operations
- Suggest appropriate strategies for CONCOR to retain its prime position in its core business in the light of increased competition

Position held: Team Member

Activities Performed: Played an important role by conducting detailed interviews with the senior officials and other stakeholders of CONCOR across India. This formed the basis for preparing CONCOR's corporate strategy, which is the largest container logistics company in India.

I. RAILWAYS:

117. Name of the assignment: Engagement of transaction advisor for setting up of *Rail Neer* packaged drinking water plants on a PPP Model at six locations.

Year: October2012 - Ongoing

Location: India

Client: Indian Railway Catering and Tourism Corporation Limited (IRCTC)

Main Project Features: *Rail Neer* is the packaged drinking water bottled and marketed by IRCTC at Indian railway stations and trains. Presently, Rail Neer meets only around 5% of the Railways' total requirement. With a view to capture a substantial share of the existing potential of the packaged drinking water market, IRCTC is exploring possibilities to manufacture and supply packaged drinking water by setting up additional bottling plants on the land available with the railways. As part of its expansion, IRCTC intends to develop packaged drinking water plants in six locations under the PPP mode. IRCTC has appointed CRIS as a transaction advisor for the same.

Position Held: Project Leader

Activities Performed: The scope of services include:

- Finalising a PPP framework for the project, identifying roles and responsibilities of each party, as well as the issues in development
- Undertaking a feasibility study for the projects
- Preparing a project information memorandum
- Conducting a pre-bid conference
- Preparing the bid documents, RFQ, RFP, and concession agreements
- Assisting in the evaluation of RFQ and RFP bids
- Assisting negotiations with the selected bidder
- Assisting in signing the concession agreement
- **4 of the locations have been awarded and financially closed**

118. Name of the assignment: Transaction & Financial Advisory Services for the Central Kalimantan Coal Railway Project

Year: 2012

Location:Jakarta, Kalimantan

Client: Indonesian Infrastructure Guarantee Fund (IIGF)

Main project features: The province of Central Kalimantan is home to some of the world's largest coking coal deposits. Yet, their exploitation has been hampered by a lack of transport infrastructure. The provincial government of Central Kalimantan (PGCK) wishes to address this infrastructure gap by procuring the construction and operation of a specialist coal rail system through a public-private partnership (PPP).

IIGF has approved the use of its internal funds to support the transaction of the Central Kalimantan Coal-Railway Project. IIGF wishes to engage the services of an investment bank to provide transaction and financial advisory services to the project and seeks to appoint qualified professional advisors to undertake and assist with the subsequent bid implementation (henceforth the transaction).

Position held: Project Leader

Activities performed:

- Manage competitive bidding, evaluation and awarding of the project in compliance with government procurement guidelines

- Manage under the supervision of IIGF, the joint planning session with bidders and miners in successfully concluding the underlying commercial arrangement relating to coal offtake for the project
- Communicate with potential financiers to assess and ensure the bankability of the project structure
- Conduct a review to fill any gaps identified in existing documentation
- As part of the negotiation with miners, calculate a fair value-based pricing for the cost of haulage using an economic approach, which subsequently forms the basis for determining the price-to-beat for the bidders`
- Coordinate the work of legal and technical advisors in updating the required information and documentations, including RFP and related agreements, as well as ensuring that all required permits and licenses have been obtained
- Facilitate communication with the bidders and other stakeholders as part of the interactive conference feature of the bidding process stipulated in the distributed RFP
- Based on inputs obtained from the bidders, assist with the follow-up including determining and recommending the necessity to incorporate such inputs
- Assist evaluation of the bids submitted and in recommending the winning bids
- Assist negotiation with preferred bidders until the signing of the contract

119. Name of the Assignment: Assistance in conducting PPP risk allocation workshop railways sector expert

Year: March 2012 to June 2012

Location: Jakarta, Indonesia

Client: Indonesia Infrastructure Guarantee Fund (IIGF)

Main project features: The IIGF's management is conducting a workshop for all stakeholders in a PPP project. The objective of this workshop to get feedback from the stakeholders regarding the risk sharing in the project. Using this feedback, IIGF will revise the risk allocation guidelines. IIGF is seeking assistance in conducting this workshop and finalising the risk allocation guidelines.

Position held: Team Member

Activities Performed:

- Prepared a context setting presentation for each sectoral discussion
- Provided necessary input on the project case (1-2 page brief write-up supplemented with draft PPP agreement) prepared by IIGF team
 - Provided the template for key provision for the PPP agreement in the selected sector, based on international best practice as to enhance understanding how the risks are allocated and drafted in the PPP agreement
- Assisted the IIGF team in managing the flow of the discussion to get the maximum value
- Shared CRIS's knowledge and experience in PPP and risk sharing in relevant infrastructure sector
- Prepare/deliver concluding remarks for each panel discussion

120. Name of the assignment: Assistance in conducting PPP risk allocation workshop road sector expert

Year: March 2012 to June 2012

Location: Jakarta, Indonesia

Client: IIGF

Main project features: The IIGF's management is conducting a workshop for all stakeholders in a PPP project. The objective of this workshop to get feedback from the stakeholders regarding the risk sharing in the project. Using this feedback, IIGF will revise the risk allocation guidelines. IIGF is seeking assistance in conducting this workshop and finalising the risk allocation guidelines.

Position held: Team Member

Activities Performed:

- Prepare a context setting presentation for each sectoral discussion
- Provide necessary input on the project case (1-2 page brief write-up supplemented with draft PPP agreement) prepared by IIGF team
- Provide template for key provision for the PPP agreement in the selected sector, based on international best practices, so as to enhance understanding how the risks are allocated and drafted in the PPP agreement
- Assist IIGF's team in managing the flow of the discussion to get the maximum value
- Share CRISIL's knowledge and experience in PPP and risk sharing in relevant infrastructure sector
- Prepare/deliver concluding remarks for each panel discussion

121. Name of the assignment: Financial appraisal of Royal Indian Rail Tours Limited (RIRTL)

Year: March 2011 to August 2012

Location: India

Client: Indian Railway Catering and Tourism Corporation (IRCTC)

Main project features: The study included:

Financial planning and modelling

- Studying DPRs at various stages of the company's formation
- Studying the working of company and its financials
- Studying RIRTL's P&L statement and its current financial position and suggesting corrective action for rationalising expenditure, keeping in mind the company's overall brand image and positioning
- Suggesting ways for rationalising income and expenditure for making RIRTL a financially viable company
- Evaluating the company's working capital needs

Marketing and positioning

- Studying the product's position in the international market and suggesting the correct positioning with respect to tariff and occupancy
- Studying marketing strategy and sales commission structure of the company and suggesting a correct strategy

Restructuring

- Studying the JV's capital structure adequacy and suggesting various possible means of restructuring capital to ensure adequacy of funding of the JV, given the current regulatory scenario

Position held: Project Leader

Activities performed: Led the study with a team of three members and was responsible for all study modules and the output quality

122. Name of the assignment: Assistance in project structuring for Kozhikode's monorail bidding process

Year: June 2011 to December 2011

Location: Kozhikode

Client: Sub-consultants to Wilbur Smith Associates (Final client: Kerala Road Fund Board)

Main project features: Wilbur Smith Associates Inc (WSA INC) has received a contract from the Kerala Road Fund Board (a state government agency), Trivandrum, for the providing consultancy services for the Kozhikode Mass Rapid Transit System project. WSA INC has appointed CRIS to assist in project structuring.

Position held: Project Leader

Activities performed: Leading the team for the execution of the assignment, which involves tasks such as:

- Review of two proposed project structures with the real estate component and with multilateral funding –
 - Assessment of risk sharing between concessionaire and the authority
 - Assessment of the practicality of multilateral funding – the possibility, related guidelines, and process with respect to three large multilateral agencies – the World Bank, ADB, and KFW
 - Analysis of overall benefits and drawbacks of each structure
 - Precedence of each structure in the domestic scenario, i.e., where else has such a structure been successfully adopted
- Recommend a suitable PPP structure for the project
- Review the available laws on urban transport projects in Kerala and suggest suitable amendments for implementing a monorail project in the city, which is a new initiative
- Define the specific role for the concessionaire and SPV for the recommended project structure
- Recommend a suitable institutional mechanism for planning, monitoring, and facilitating the project during the implementation stage
- Modify any bid documents, concession agreements, or other project deliverables to incorporate the suggested project structure. These modifications will only reflect the commercial value and risk sharing principles and the subcontractor need not vet the documents legally
- Interact with/make presentations to KRFB only on the deliverables by the subcontractor

J. FINANCIAL SERVICES:

123. Name of Assignment: Options study for Railways of India Development Fund

Year: November 2015-Ongoing

Location: Delhi, India

Client: The World Bank

Main Project Features: The World Bank wishes to assist the Ministry of Railways to establish a Railways of India Development Fund (RIDF) to mobilise non-government funding for investments in railway infrastructure. The purpose of raising these funds is to undertake modernization and capacity augmentation projects, which will enhanced revenue and cash flow generation for the Railways.

The RIDF will establish a long-term debt and equity financing mechanism for railway infrastructure investment to complement established channels of rail financing such as internally generated funds, budget allocations and leasing through Indian Railway Finance Corporation.

Position Held: Project Leader

Activities Performed:

- Phase 1: Analysis and Fund Structuring
 - Legal analysis
 - Investment pipeline
 - Funds structure
 - Financial analysis
- Phase 2: Market testing and implementation
 - Market testing
 - Operations manual
- Phase 3: Impact assessment and communication plan

124. **Name of the assignment:** Developing a project completion risk guarantee facility for India

Year: August 2014 to December 2014

Location:India

Client: Asian Development Bank

Main project features: The objective of the proposed facility will be to provide guarantees to infrastructure project special purpose vehicles (SPVs) set up through PPP mode, for risks at the pre-completion stage which are outside the control of the SPVs. Risks would typically include with respect to delays in land acquisition, environment and forest clearances, state government clearances, aggregation risk (delays due to non-availability of raw materials), law and order etc. These delays typically arise on account of breaches in the concession agreement by the public-sector authorities. In the event of delays on account of the above and consequent risks to project cash flows, the proposed PCRGI would pay off the lenders that are exposed to the project. The PCRGI would charge a guarantee fee. CRIS has been appointed by the Asian Development Bank to design such a facility.

Position Held: Project Leader

Activities Performed:

CRIS is advising in the following areas:

- Market assessment – Establish relevance of the PCRGI
- Study similar facilities
- Develop business plan

125. **Name of the Assignment:** Validation of strategy developed for entering into rural finance business

Year: August 2014 to January 2015

Location:India

Client:Tata Capital Financial Services Limited

Main project features: Tata Capital Financial Services Ltd (TCFSL) plans to foray into the rural finance business. The entity designed a blueprint detailing out product offerings, sourcing mix, business model and strategy, overview of regulations, etc.

The entity wished to engage a third party for a quick-fire validation (over 2 weeks) of this blueprint. CRIS is pleased to present its scope of work based on initial discussions had with the client.

Position held: Project Leader

Activities performed:

Module 1: Rural financing strategy validation

- Context setting – Interactions with management to understand objectives, vision of the rural finance business
- Review of the blueprint developed by TCFSL on its rural finance business. The review was conducted to validate the following aspects of the strategy as provided by TCFSL in its blueprint:
 - The defined lines of business were segmented, viz. tractor financing, auto financing, agricultural and allied activities (including dairy and warehouse receipt financing) and last-mile activities
 - Market potential of business lines were estimated
 - Competition analysis
 - Risks analysis
 - Strategy evaluation
 - Positioning
 - Operating model
 - Risk mitigation
 - Pricing
 - The validation exercise would be carried out using following methods:
 - Interaction with in-house sector experts
 - Secondary research, including internal databases
 - The scope of validation would be more in detail for the agricultural and allied activities vertical and comparatively broad-based for the other segments

Module 2: Evaluation of two states (including Uttar Pradesh) with the aim of analyzing prospects from TCFSL's perspective

Module 3: Analysis of the warehouse receipt and dairy financing verticals

Work on modules 2 and 3 were pre-dominantly based on secondary research from available information sources and very limited primary research through telephonic interactions.

126. **Name of the Assignment:** Valuation of assets for an NBFC

Year: May 2014 – September 2014

Location:India

Client:Confidential

Main project features: A leading NBFC appointed CRIS as a consultant for independent third party valuation of security for 11 assets in its books

Position Held: Project Leader

Activities Performed:

The scope of work included valuation of security for 11 assets in the books of the NBFC. The said security *inter-alia* consists of charge on movables, immovable, current assets, pledge of shares of borrower, Corporate/Personal Guarantee of Promoter Company/Promoter etc.

127. **Name of the assignment:** Development of a Bond Guarantee Fund for India (BGFI)

Year: July 2014 to September 2015

Location: India

Client: Asian Development Bank

Main project features:

The draft twelfth Five-Year Plan (2012 to 2017) document has set a GDP growth target of around 8% over the period. Investments, which are crucial to achieve growth, are forecasted to reach 34.2% of the GDP during the period with infrastructure investments alone expected to amount to \$1 trillion. A major portion of these investments were envisaged to be carried out by the private sector. It was thus important to understand the potential sources of finance for these investments.

Banks have been the dominant source of finance in the country, with domestic credit forming more than 74% of the GDP. However, in recent years, the banking sector has been facing stress, which need for alternative sources of finance. Corporate bond issuances have grown from Rs 40,000 crore in 2003-04 to around Rs 351,000 crore in 2012-13 – the market suffers from concentration issues with regards to the kind of issuers and rating of issuances. About 90% of issuances had a rating of AA- and above, signifying an almost non-existent market for low-rating issues. Therefore, low-rated entities are not able to source their funds from the bond market, signifying a gap.

CRIS was appointed by ADB to design a Bond Guarantee Fund, which would provide credit enhancement to bond issuances in the infrastructure and non-infrastructure sector through a wide range of credit enhancement mechanisms, so that they can access the bond market.

Position held: Project Leader

Activities performed:

CRIS is advising in the following areas:

- Market assessment – Establish relevance of the BGFI
- Develop a detailed business model
- Design broad institutional frameworks and structures
- Disseminate knowledge through roadshows and seminars

128. **Name of the Assignment:** Bid advisory services for Mauritius Light Rapid Transit Project

Year: May 2014 to August 2014

Location: Mauritius

Client: AFCONS Infrastructure Ltd

Main project features:

The client was in the process of submitting a PPP bid to the Government of Mauritius for the Mauritius Light-Rapid Transit (MLRT) system project. To keep pace with the expansion in Mauritius' economy, increased private vehicle use and traffic congestion in Port Louis and surrounding areas, the Government of Mauritius (GoM) decided to explore the provision of a light rapid transit system through assistance from the Government of Singapore in 2010.

As part of this project, CRIS is helping Afcons in completing certain non-technical sections of the documentation as prescribed by the internal matrix under the RFP

Position Held:Project leader

Activities performed:

The scope of work for this assignment included the following two modules:

Module 1: Whole of Life cost. This included:

- Financial model incorporating completed MLTA financial model pro-forma worksheets P1 to P6 for Scenario 1, where the minimum services is delivered (base case) and Scenario 2, where maximum services are delivered.
-
- Capital and operating costs were estimated based on the inputs provided by AFCONS
- Inputs will be sought from Afcons to understand prevailing scenario of financing arrangements of similar projects being undertaken by Afcons.
- Manpower costs shall be on basis of staffing plan which shall be undertaken with inputs from Afcons
- All the other cost may be clubbed or taken on lump sum basis as available
- Data book supporting the financial model.
- Copy of letter from financial model auditor certifying that the financial model has been audited and that all calculations are correct.
- Overview of the taxation issues affecting the Project (the exercise will be limited to known impact of taxation on financials)

Module 2: Overall Vision and Partnership

- A summary of Afcons' overall vision, plan and approach to meet the GoM's strategic objectives as outlined in scope.
- An overview of the key features of Afcons' Proposal, including its conforming proposal and any alternative proposal(s) (if applicable). The alternative proposal shall be provided by Afcons
- A description of Afcons consortium and structure, including full details of the proposed management structure, key management personnel and the relationships between the management team.
- A description of how Afcons intends to address continuity of management personnel through and between the various phases of the project. The description will be a brief comment on the structure
- A description of how Afcons intends to address accountabilities and responsibilities of the management personnel.
- A description of how Afcons intends to address governance and issue resolution arrangements
- Details of how Afcons' proposal will support the development of Mauritius
- A statement of Afcons' approach to establishing and maintaining a partnership with the key Project partners identified in section 3 of the RFP Overview (Volume 1), including the requirements listed in paragraph I of Returnable T.O.

- A summary of the key scope, technical and operational features of Afcons' proposal and how the GoM's project vision and objectives will be met. The inputs in documentary form for the same shall be provided by AFCONS.
- Project initiation and completion

129. **Name of the assignment:** Operational manual improvement for IIGF

Year: June 2014 to December 2014

Location: Jakarta

Client: IIGF

Main project features:

CRIS was appointed by IIGF for improvement of the 2011 operational manual for IIGF as per the changes in the regulations and legislations.

Position Held: Project Leader

Activities performed:

The scope of work for this assignment included the following modules:

- Operational manual
- Marketing toolkit
- Underwriting policy

130. **Name of the assignment:** Review of progress of recommendations proposed to strengthen Mutual Benefit Trust structure.

Year: February 2014 to April 2014

Location: India

Client: Sarvam Mutual Benefit Trust, Kancheepuram

Main project features:

Sarvam Mutual Benefit Trust Kancheepuram ("SMBT", "Sarvam") is a private trust, which serves as an apex trust and investment advisor to 119 mutual benefit trusts (MBTs) located in Tamil Nadu, Karnataka and Madhya Pradesh; these MBTs serve as aggregators of Self Help Groups (SHGs), which are informal groups of 12-20 women in particular regions. Sarvam is one of the shareholders of Belstar Investment Finance Pvt. Ltd (BIFPL). BIFPL had appointed CRIS in 2012 to undertake an independent evaluation of the extant governance structure then and provide its recommendations. CRIS was therefore appointed by Sarvam to review the progress against its recommendations.

Position Held: Project Leader

Activities Performed: The following activities were undertaken for the execution of this assignment

- Context setting and stock-taking of progress of implementation of recommendations
- Interactions with sample set of MBTs
- Observations and recommendations

131. **Name of the assignment:** market study on movables financing in India

Year: May 2013 to August 2014

Location: India

Client: International Finance Corporation (IFC)

Main project features: The IFC, the private sector development arm of the World Bank group, has played an active role in enhancing access to movables finance for the Micro Small Medium Enterprise

(MSME) sector across developing countries. In India, the use of movables as a collateral is relatively very limited. As a result, many MSMEs are effectively excluded from the formal financial system. Hence, IFC was keen to evaluate the potential for financing MSMEs purely against movables as collateral.

Accordingly, IFC mandated CRIS to study the current status of movables finance to the MSME sector in India and arrive at the market potential for financing MSMEs purely against movables.

Position held: Project Leader

Activities Performed: The study had three key objectives:

- Estimating current level of finance available to MSMEs in India, against movable collateral:
- Types of collateral used
- Total level of financing (volumes and number of MSMEs) that receive finance against movables
- Level of finance to women owned MSMEs against movables
- Level of credit extended in India against movable collaterals to global benchmarks
- Identifying models of movable financing that have been used by financial institutions including:
 - Successful models and their potential for replicability
 - Unsuccessful models and reasons for their failure
- Analysing size of the market for financing against movable collateral that can be potentially tapped by the financial sector
 - Provide an estimate of percentage of credit requests that are rejected by financial institutions
 - Understand the current market perceptions about hurdles to increasing the level of movables finance
 - Any policy or enabling environmental constraints that need to be considered

132. Name of the assignment: Development of a business dashboard for Peerless Developers Limited's (PDL's) management

Year: July 2013 to November 2013

Location: India

Client: Peerless Developers Ltd.

Main Project Features: CRIS's study laid down the initial steps for Peerless Developers Ltd (PDL) to progress towards achieving its long-term vision of becoming a 'financial products superstore'. CRIS gave a concise and holistic overview of the entity's operations in terms of its history, future goals, and key steps being adopted to achieve these goals.

Position held: Project Leader

Activities performed:

- Review of operational plan and business key performance indicators (KPIs)
 - Interaction with PDL's Management to understand the objectives and initiatives under the operational plan
 - For each component in the operational plan review KPI's (baseline and targets)
- Drill-down of business KPIs to input-output measures
 - Interact with the management to understand the relative importance of each KPI
 - Interact to understand the roadmap in terms of input-output measures planned by the management
 - Develop business dashboards

133. Name of the assignment: Fee-based services strategy, business plan, and operationalisation

Year: August 2013 to April 2014

Location: Indonesia

Client: PT Sarana Multi Infrastruktur (Persero) Tbk

Main Project Features: The client wanted to emerge as a leading financing institution that focuses on lending to infrastructure projects. CRIS has recently undertaken a comprehensive study for PT SMI, wherein the following were developed:

- Product library manual
- Credit policy and equity investment policy and
- Process manual

PT SMI appointed CRIS to undertake a strategy development for its advisory services division including development of a business plan and its operationalization.

Position held: Project Leader

Activities performed:

- Assessment of PT SMI's existing advisory services business and its goals and internal capabilities
 - Discuss and review existing structure with the senior management to interpret PT SMI's mission of accelerating infrastructure development
 - Summarising strengths and weaknesses for offering advisory services
- Market assessment and international benchmarking of advisory services
 - Study the prevailing advisory services environment in Indonesia and comparative analysis with the international business model
 - Summarising opportunities and threats for offering advisory services
- Articulation of advisory services' role and objectives for PT SMI
 - Discussed SWOT analysis and broad advisory services that SMI may offer.
- Developing business plan for PT SMI's advisory services
- Operationalisation of advisory services

134. **Name of the assignment: Advisory services for product library development and resultant policies, processes and manuals**

Year: March 2013 to July 2013

Location: Indonesia

Client: PT Sarana Multi Infrastruktur (Persero) Tbk

Main Project Features: PT Sarana Multi Infrastruktur (Persero) was established in 2009 to accelerate infrastructure development in Indonesia. PT SMI focused on funding the commercially feasible infrastructure projects through debt, equity, and mezzanine instruments to address the constraints on the flow of investment in infrastructure. In order to provide financing to infrastructure projects in Indonesia on reasonable terms, the PT SMI management was looking to develop an effective product library and having a suitable enabling environment in terms of policies, processes, and manuals to support the same.

PT SMI thus appointed CRIS, India, to provide advisory services in this regard.

Position held: Project Leader

Activities performed:

- Review and improvement of product library

- Development of financing and investment policies
- Review and improvement of business processes
- Development of credit and investment process manuals
- Conducting a workshop on effective implementation

135. Name of the Assignment: Evaluation of Business Plan for NEIA Trust

Year: December 2012 to June 2013

Location: India

Client: Export Credit Guarantee Corporation (ECGC) of India

Main Project Features: The National Export Insurance Account (NEIA) Trust is a public trust set up in 2006 by the Government of India and administered by the ECGC. The trust provides export credit insurance support to facilitate medium- and long-term exports to difficult markets. The trust also provides covers to banks for buyer's credit transactions, which facilitates foreign buyers to pay for project exports from India. ECGC had appointed CRIS to evaluate its business plan.

Position held: Project Leader

Activities Performed: The scope of work for the current study included:

- Performance evaluation of the NEIA Trust, since its inception in 2006
- Scope of expanding the NEIA Trust's coverage beyond medium- and long-term business
- Specific measures to boost the exports to specified countries to correct India's trade imbalance, through the trust.

136. Name of the assignment: Evaluation of business plan for ECGC

Year: December 2012 to July 2013

Location: India

Client: ECGC

Main Project Features: Export Credit Guarantee Corporation of India Ltd (ECGC), established in 1957, is a leading public sector enterprise providing export credit insurance facilities to exporters and banks in India. ECGC is the seventh-largest credit insurer of the world in terms of coverage of national exports. ECGC seeks to improve the competitive capacity of Indian exporters by giving them credit insurance covers comparable to those available to their competitors from most other countries. The maximum liability that the company can underwrite at any point of time is Rs 1, lakh crore. It had appointed CRIS to evaluate its business plan.

Position held: Project Leader

Activities performed: The scope of work for the current study included:

- Operational and market assessment for existing business
- Regulatory assessment
- International benchmarking
- Developing the future market potential
- Developing a road map for increasing ECGC's share in coverage of India's exports
- Developing financial projections

137. Name of the assignment: Review and improvement of credit administration and credit monitoring processes

Year: October 2012 to May 2013

Location: India

Client: Tata Capital Limited

Main project features: The client, a subsidiary of Tata Sons Limited, is registered with the Reserve Bank of India as a systemically important non-deposit-accepting core investment company and offers through itself and its subsidiaries fund and fee-based financial services to its customers. Tata Capital has a corporate and consumer finance division. The corporate finance division has commercial and infrastructure finance.

Tata Capital was keen to undertake a review of the prevailing processes in its credit administration and credit monitoring departments for its corporate finance and infrastructure finance divisions. The review was meant to diagnose the existing processes and organisation structure in these departments, identify areas of concern, and suggest improvements that minimise risks as well as improve the effectiveness and efficiencies in these processes.

Tata Capital therefore appointed CRIS to undertake this study.

Position held: Project Leader

Activities Performed: The scope of work for the current study included:

- Context setting and an as-is process mapping
 - Undertake prioritization of objectives that defines Tata Capital's process outcomes
 - Reviewing the prevailing regulatory framework and possible impact on processes
 - Reviewing the existing policies at Tata Capital related to credit administration and monitoring
 - Mapping detailed processes and sub-processes in each department – activities, resources allocated, turnaround time, maker/checker systems, controls/approvals, MIS, extent of templatisation and standardisation of documents
- Critical evaluation and diagnosis of process
- Identifying gaps/areas of improvement – in terms of activities, built-in operational controls, extent of resource allocation and organization structure, documentation and MIS, prevailing level of automation
- Recommending process improvements
 - Listing out various recommendations – in terms of activities, documentation, MIS, organization structure, etc.
 - Developing a pain-gain framework for implementation of recommendations

138. **Name of the assignment: Monitoring and benchmarking of subsidiaries**

Year: September 2012 to July 2013

Location: India

Client: Peerless General Finance & Investment Co. Ltd.

Main Project Features: Peerless General Finance & Investment Co. Ltd. (PGFI) is a registered residuary NBFC, incorporated in 1932 with a pan-India presence. PGFI has diverse business interests in varied sectors like financial services, hospitality, healthcare, and real estate, which are housed in several subsidiaries. PGFI is looking to formalize a suitable framework to monitor and benchmark operations of these subsidiaries on a consistent basis. It has thus appointed CRIS to provide advisory services.

Position held: Project Leader

Activities performed: The scope of work for the current study included:

- Development of framework for monitoring and benchmarking for five subsidiaries
 - Studying each subsidiary's business model, areas of operation, and performance

- Understanding management's (of the PGFI and each subsidiary's) corporate vision, mission and operating philosophy in order to gauge the management's aspirations and constraints within which they intend to function
- Identifying parameters (industry, competition, and standalone) to be included in the framework based on data available in the public domain and within CRISIL
- Implementation of framework for four cycles
 - Collect data for monitoring of subsidiaries for various identified parameters
 - Collect data for benchmarking of subsidiaries
 - Populate the framework with the data collected
 - Identify data gaps and possible resolution
 - Finalize the framework

139. Name of the Assignment: Advisory services for NBFC business

Year: October 2011 to May 2014

Location: India

Client: Tata Capital Limited; **Position held:** Project Leader

Main Project Features: Tata Capital Limited (TCL), a subsidiary of Tata Sons Ltd., is a non-deposit-accepting NBFC. It has a presence in various fund-based and fee-based businesses. The fund-based business has two –divisions - corporate and consumer finance. The Infrastructure Finance Division (IFD) is a sub-division of the Corporate Finance Division.

TCL started the IFD in 2007 and has a presence in construction equipment finance, working capital finance, and promoter financing. It is not very active in the project finance segment yet. TCL is keen to develop a strategic plan for IFD and has appointed CRIS to provide advisory services

Activities Performed: CRIS is advising in the following areas:

- Assessment of current operations of the IFD
- Market assessment of infrastructure finance in India
- Competitor profiling of infrastructure finance NBFCs in India
- Overview of NBFC types and regulatory environment
- Strategic approach plan for the infrastructure finance business
 - Understanding the objectives and ideology of TCL
 - Identifying Synergies within Tata group, if any
 - Developing sector focus strategy
 - Analysing and identifying product types to be offered
 - Developing a timeline-based growth strategy in the light of market opportunities and resource availability
 - Assessing additional capital and human resource for the growth strategy
- Financial projections and risk analysis
- Annual reviews of strategic plans

140. Name of the Assignment: Market assessment of Indian debt securities market and implications on IRDA's investment strategy

Year: December 2011 to February 2013

Location: India

Client: Insurance Regulatory and Development Authority (IRDA); **Position Held:** Project Leader

Main Project Features: The Insurance Regulatory and Development Authority (IRDA) is a national agency of the Government of India, based in Hyderabad. It was formed by an act of Indian Parliament known as the IRDA Act (1999). IRDA is the regulator for life and general insurance sectors in India.

IRDA was interested in conducting a market assessment of the Indian debt-securities market with the objective of understanding the implications of the demand-supply estimates on IRDA's investment regulations.

Activities Performed: CRIS advised in the following areas:

- Overview of Indian insurance industry and comparison to global industry
- Overview of existing regulations governing the Indian insurance industry
- Demand assessment of government and corporate debt securities by banks, the insurance industry – life and general insurance, mutual fund industry, pension funds, Provident Fund (EPFO), Reserve Bank of India (RBI), and foreign institutional investors
- Supply assessment of government and corporate debt securities
- Assessment of the demand-supply situation
- Implications of the demand-supply situation on IRDA regulations for both government and infrastructure corporate debt securities

141. Name of the assignment: Capacity development for Indian Infrastructure Finance Company Ltd

Year: May 2011 to October 2011

Location: India

Client: Asian Development Bank; **Position held:** Team Member

Main Project Features: It is envisaged that India would require around \$1 trillion of infrastructure investments during the XIIth Five-Year Plan. The funding for infrastructure investment has predominantly been provided by the banking sector thus far. But in view of the fact that infrastructure investments grew to account for about 14% of the asset book of the Indian banking sector (in 2011) and the asset-liability mismatch (ALM) risk posed by the lengthening maturities of loans (required by infrastructure projects), it is possible that banks might not be able to continue funding infrastructure at an increasing scale.

Therefore, looking for alternative sources of infrastructure finance is necessary. The most promising of these alternatives is the bond market. The development of the non-sovereign bond market in India has been tepid in part because of the investment norms of large bond investors (such as insurance and provident funds), who have an inevitability threshold of 'AA' or higher. The thresholds shut out bond market access to most infrastructure projects, which tend to be placed in the BBB category and thereabouts. The Economic Advisory Council to the Prime Minister (PMEAC) recommended that the India Infrastructure Finance Company Limited (IIFCL) design a **credit enhancement scheme** that could be used to enhance the ratings of the infrastructure project bonds such that the projects are able to access finance from a larger number of institutional long-term investors like insurance and provident funds.

IIFCL employed CRISIL Infrastructure Advisory to design such a scheme

Activities Performed: CRIS advised in the following areas:

- *Market assessment*

- Brief overview of the infrastructure financing situation in India, the key constraints and issues in present financing
- Need for alternative sources of financing such as the bond market, reasons behind the undeveloped bond market, and ways in which credit enhancement would help deepen the same
- Overview of the various modes of credit enhancement and how PCG is the suitable instrument for achieving the objectives of the credit enhancement scheme for IIFCL
- *Overview of the PCG product*
 - Detailed description of PCG and its various features, including the benefits and challenges that could be faced by the product in the market
 - Examples of PCG transactions in India and abroad
- *PCG structure and mechanics*
 - Lay out parameters for such a product in terms of project eligibility criteria, PCG parameters and pricing
 - Financial analysis of the product to ascertain its viability
- *Designing the PCG scheme*
 - Laying out the PCG scheme document
 - Overview of legal and taxation issues to be addressed in such a scheme
 - Stakeholder consultations

142. Name of the Project: Feasibility study for supporting Indian Investment abroad in strategic agricultural activities

Year: May 2010 to December 2010

Location: Africa and Latin America

Client: Export-Import Bank of India

Main Project Features: Export-Import Bank of India is a wholly government-owned financial institution, established under an Act of Parliament, to promote, facilitate and finance India's international trade and investment. The bank offers a diverse range of products and services, including a variety of export credit facilities, and finance for export-oriented units for their ventures in India and abroad. Many developing nations, especially Africa and Latin America, have been liberalizing foreign investment in agriculture cultivation to enhance the potential of job creation in rural areas and thereby reduce poverty, and tackling food scarcity. Many emerging nations, either directly or through state-owned entities, or under public-private-partnership, are in negotiations / or have close deals on leasing / purchase of lands for cultivation abroad. Indian companies have also been investing abroad for agriculture cultivation through acquisition of large tracts of land. It is opined that encouraging Indian investors in strategic agricultural cultivation would be beneficial for India, as the country depends on import of many agricultural items.

In the above context, EXIM Bank intended to undertake a feasibility study to gauge the potential for supporting Indian investment abroad in strategic agricultural activities.

Position Held: Project Leader

Activities Performed: CRIS was involved in the following activities:

- Identification of potential for financing Indian investment abroad in strategic agricultural activities;

- Identification of strategic agricultural products (to meet the challenges of food and energy security);
- Identification of target countries based on the potential for producing the identified products, and the policy framework in the target countries facilitating foreign investment inflows into agriculture sector, in general and land acquisition in particular.
- Identification of strategic segments (such as production, post-production, R&D, processing, agri-biotechnology, cold-chain, institutional cooperation, rural infrastructure) to undertake capacity building activities in target countries.
- Identification of types of assistance that may be needed (whether funded / non-funded facility, whether term loan, mezzanine financing, equity participation; whether assistance towards capital expenditure or working capital);
- Identifying feasibility / challenges in creation and enforceability of security, suggested credit risk management measures.
- Identifying suitable monitoring mechanisms, including safeguards that may be required, if financial assistance could be considered.

K. PPP & OTHERS

143. Name of the assignment: Feasibility study for setting up an industrial unit

Year: March 2013 to August 2013

Location: India

Client: Confidential; **Position held:** Project Leader

Main project features: Location-based feasibility study

Activities Performed: The scope of work for the study included:

- Overview of the product-specific industry
- Identification of four potential states for setting up the plant
- Identification of a suitable location and development of a decision matrix for the client

144. Name of the assignment: IDC TA 7152 IND PPP Pilot Projects Initiative IDCA food storage infrastructure

Year: February 2013 – December 2015

Location: India

Client: Asian Development Bank; **Position held:** Project Leader

Main Project Features: The Asian Development Bank (ADB) is supporting the Government of India (GoI) with technical assistance (TA) for mainstreaming PPPs by developing pilot projects in challenging sectors, selected jointly between ADB and the Department of Economic Affairs (DEA), the executing agency for the TA.

Food storage infrastructure is a critical need in the country, since it is estimated that post-harvest losses of food-grains are of the order of 10%, which constitutes a national waste. Lack of covered storage forces state agencies to store food-grains in the open, and with over 25% of storage being in the open¹, the situation is far from satisfactory. In the circumstances, in February 2012, an empowered group of ministers (EGOM) decided to create substantial storage capacity of about two

million tons through construction of silos. Since storage infrastructure comes under the viability gap fund scheme of GoI, it is possible to create such storage infrastructure through PPP.

Activities Performed: The scope of work for this assignment shall include:

- Provide key guiding factors to enable development of a model project development agreement (PDA) specific to the food storage sector and successfully undertake the transaction to establish a scalable/replicable model for food storage
- The PDA will incorporate contract provisions and guidelines as a reference for food storage Public Private Partnership (PPP) contracts in India. The aim of the document is to:
 - Enshrine both domestic and international best practice in the development of PPP contracts in the sector, promoting, bankability. "
 - Build key contract data points applicable for the food storage sector
 - Promote a common approach to risk allocation in PPP contracts in the food storage sector
 - Reducing the time and costs of developing PPP contracts and laying down reference guidelines
 - Pilot 1-2 transactions identified by DEA in collaboration with FCI, including stakeholder discussions and management of procurement process ending with successful signing of the concession agreement.
 - Advise FCI in putting in place a contract management unit to monitor the projects, including preparation of the indicators for monitoring and the recommended periodicity of the indicators (this should include critical clauses of the Concession Agreement to ensure timely implementation of the project and KPIs to ensure that the output is at the prescribed levels).
 - Assist FCI in preparing the contract document for selection of independent engineer for the PPP projects

145. Name of the assignment: PPP institutional reform capacity building and pipeline development support

Year: November 2013 to September 2014

Location: Sierra Leone

Client: The World Bank; **Position held:** Project Leader

Main Project Features: The Government of Sierra Leone (GoSL) planned the use of public-private partnerships (PPPs) as a mechanism for procuring and financing infrastructure projects and services in the public sector. To achieve this goal, the World Bank Group (WBG) is supporting the recently established PPP Unit under the Office of the President, GoSL.

The WBG engaged the services of CRIS as consultant to help in the initial stages of PPP pipeline development

Activities performed:

- PPP Project development advisory support focusing on assessing and developing the GOSLs current pipeline of PPP projects:
- Reviewed and discussed the current pipeline of projects in the energy and other infrastructure sectors with the GOSL, to assess the feasibility of private sector participation as well as provide an assessment of the technical, financial, operational and management situation (including pros and cons) of potential transactions
- Identification of potential greenfield projects:
- The report also included recommendations on feasible greenfield projects for the short and medium-term
- A short options paper on possible GoSL's models of engaging concessionaires on energy

- Drafting of a short options paper which will outline the possible GoSL models of engaging concessionaries on government acting as an off taker from concessionaires that build power plants, government having equity participation in SPV vehicles for generation, government as an owner and selling to companies

146. Name of the assignment: Assistance in operationalising Namibia's PPP Policy

Year: June 2013 – July 2014

Location: Namibia

Client: Ministry of Finance Namibia; **Position Held:** Project Leader

Main Project Features: The Government of Namibia has taken the first steps towards attracting private investments in the areas of economic, social, municipal, or social assets by way of developing a national PPP policy, as per whose provisions, the Ministry of Finance (MoF) is expected to take a leading role in enablement and approval of PPP projects in the country. With this backdrop, MoF appointed CRIS to assist them on various activities related to operationalising the PPP policy.

Activities Performed: The key activities envisaged under this engagement are:

- Legislative review and law creation to enable PPPs
- Development of a 'review and approval process' for annuity PPP projects to ensure fiscal soundness
- Preparation of guidance material for all stages of the PPP process
- Assistance in organization of the PPP unit under the MoF
- Capacity building: design and delivery of agreed PPPs related training programs
- Identification of two pilot PPP projects, which can be further taken up for development and PPP procurement
- Stakeholder consultations
- Preparatory works for approval of the PPP Bill
- Taking the Bill through various stages of approval process

147. Name of the Assignment: PPP Pipeline and Viability Gap Mechanism Assessment

Year: August 2012

Location: Ghana

Client: World Bank; **Position Held:** Project Leader

Main Project Features: The Ghanaian and Kenyan governments have proposed 11-12 projects that constitute the 'first mover projects' (FMP) under their respective PPP policies. The initial assessment of the suitability of these projects for PPP financing is being carried out. On the basis of the financing requirement assessments (equity, debt, guarantees, etc., and other government support), the bank will be: (i) providing guidance to the governments on how best to structure financing instruments, including VGM options, and (ii) providing funding support to some of these options under Phase II of the program.

Activities Performed: The scope of services provided by CRIS included:

- Preparing the preliminary VGM profile for each of the FMPs in Ghana and Kenya.

- Compilation of international approaches\methodologies to VGM, pros and cons, and performance to date (e.g., IUK, South Africa, India, Chile, Brazil; Malaysia, South Korea, New Zealand, Canada) building also on work done to date by World Bank;
- Recommendations on design options for as to the most efficient approaches for a VGM with specific attention to ongoing integrated financial management information systems (IFMIS) underway in the respective countries to insure effective VGM integration into overall budget systems to allow for effective\efficient recording of VGM allocations in main estimates and regular reporting.
- Participation in the September 10-21 mission to Ghana and Kenya;
- Assistance on organization of South-South Practitioner Workshop to be hosted in Nairobi, including background presentations

148. Name of the assignment: Consultancy services for TA for development of bus terminal complex at Trichy under the PPP model

Year: December 2009 to June 2010

Location: India

Client: State Express Transport Corporation (Tamil Nadu Division-Ii) Ltd.; **Position held:** Project Leader

Main project features: The client (earlier named Thiruvalluvar Transport Corporation) operates long-distance buses exceeding 250 km throughout the state of Tamil Nadu, linking all important capital cities and adjoining states including Andhra Pradesh, Kerala, Karnataka and the Union Territory of Pondicherry. To decongest the existing bus terminal and improve the quality of passenger services, the corporation initiated a project to develop a new bus terminal on a 0.6 acre plot under its possession exclusively to serve SETC buses on a PPP basis. As a part of the same, it selected CRIS, through a competitive bidding process, to avail feasibility and transaction advisory services for the project.

Activities performed:

- Pre-feasibility study for bus terminal and related commercial developments including traffic studies, market assessment, market sounding, financial analysis, and feasibility study
- Conceptual layout for the development
- Drafting of bid documents including the RFP and concession agreement
- Assistance to the client in conducting pre-bid meeting and issuing clarifications
- Selecting of an appropriate developer through a transparent and competitive bidding process
- Selection of an independent engineer to monitor the progress of development

149. Name of the assignment: Developing Tool Kit(s) for Improving PPP Decision-Making Processes

Year: August 2009 to July 2011

Location: New Delhi, India

Client: The World Bank; **Position held:** Project Leader

Main Project Features: The objective of the proposed assignment was to develop a web-based online 'PPP Toolkit' to facilitate identification, assessment, development, procurement, and monitoring of public private partnership (PPP) projects, with a focus on 4-5 selected infrastructure

sectors— roads, ports, urban transport like bus rapid transit system (BRTS), water supply and sanitation, and solid waste management.

These toolkits—both generic as well as sector-specific variants—provided relevant, practical, and best practice guidance to officials at the national, state, and city/local governments, state-owned enterprises, and other agencies responsible for developing and monitoring PPPs in the selected sectors. Accordingly, the toolkits had been developed based on an assessment of performance of a representative set of PPPs that were already awarded and were in operation, so as to build upon the lessons from the past experience. Additionally, they incorporated international best practices by drawing upon toolkits that were already developed by the World Bank/Public-Private Infrastructure Advisory Facility (PPIAF)/other agencies. These toolkits were expected to be particularly useful for lagging states and regions with huge infrastructure deficits and low capacity to conceptualize and develop PPP projects. The toolkit also contained case studies of 15 landmark PPP projects.

Activities performed:

- Reviewed selected PPP projects across identified sectors as case studies to identify learnings that could be used for development of toolkits
- Provided necessary inputs for the development of web-based online toolkits
- Prepared MS Excel models
- Prepared user manuals, and organize and conduct stakeholders' meetings

150. **Name of the assignment:** Conducting of pre-feasibility study for development of integrated transport hub in 25-acre land parcel at Byappanahalli (NGEF Land)

Year: July 2010 - August 2010

Location: India

Client: Karnataka State Road Transport Corporation; **Position Held:** Project Leader

Main Project Features: Karnataka State Road Transport Corporation (KSRTC) and Bangalore Metropolitan Transport Corporation (BMTCL) jointly proposed to develop a state-of-the-art integrated bus terminal in a 25-acre plot obtained from NGEF at Byappanahalli, for operating inter-city buses (those entering Bengaluru through NH4 and NH7 and outer ring road) as well as city bus services. The bus terminal shall a) provide world-class passenger facilities and other transport related facilities and b) act as an inter-modal transit centre by feeding and catering to passengers using other transit modes including the inter-city buses, city buses, metro rail station, and railway station that are in close proximity.

The scope of work included:

- Traffic studies and financial feasibility assessment - Understanding the current traffic levels, evaluate the past trends, project traffic based on new developments and plans, arrive at the block cost estimates of the project, prepare a financial model, and assess key financial return ratios
- Preparation of circulation plans – Evaluating the existing connectivity from NH to the location, issues of congestion, and suggest 2-3 alternative circulation plans to decongest the area and enable hassle-free circulation for short and long terms
- Preparation of concept designs for the entire development and connectivity plans – Developing a concept plan for the integrated transport hub by taking into consideration

	options for PPP development, and making it an integrated commercial, transport and parking hub catering to bus terminals of KSRTC, BMTC and parking requirements of the adjacent metro rail terminal station • Study the inter-modal connectivity options – Considering the options for effective connectivity between various modes of transport including local and outstation bus terminals, parking facilities, and the adjoining metro-rail terminal, while preparing the plans • Suggesting plans for development and project structuring including evaluation of viability of the project to be developed on PPP basis Activities Performed: • Prepared the framework of the study to be conducted by the team • Participated in all the client interactions and presentations • Reviewed the quality of output key • Acted as an interface for the client
	B.